



<Customer Full Name> Database Service on Exadata Cloud@Customer

Solution Definition

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Document Control

Guide:

The first chapter of the document describes the metadata for the document. Such as versioning and team members.

1.1 Version Control

Guide:

A section describing the versions of this document and its changes.

Example:

Version	Authors	Date	Comments
1.0	Author name	10 th Aug 2026	First release

1.2 Team

Guide:

A section describing the team.

Example:

Name	Email	Role	Company
Author #1	example@example.com	DBP Account Cloud Engineer, Author	Oracle
Author #2	example@example.com	Database-C@C BB, Contributor	Oracle

1.3 Abbreviations List

ACME	A Company that Makes Everything (Global Bank in this document)
OCI	Oracle Cloud Infrastructure
VCN	Virtual Cloud Network
VM	Virtual Machine
AD	Availability Domain
FD	Fault Domain
DR	Disaster Recovery
DBaaS	Database as a service
MAA	Maximum Availability Architecture
RAC	Real Application Clusters
RMAN	Oracle Recovery Manager
OKV	Oracle Key Vault
DNS	Domain Name Server
PKI	Public Key Infrastructure
BOM	Bill of Material

AES	Advanced Encryption Standard
DMZ	de-militarized zone
ExaDB-C@C	Exadata Database Service on Cloud@Customer
ZDLRA	Zero Data Loss Recovery Appliance
RPO	Recovery Point Objective
RTO	Recovery Time Objective

1.4 Document Purpose

Guide:

Describe the purpose of this document and the Oracle-specific terminology, specifically around 'Workload'.

Example:

This document provides a high-level solution definition for the Oracle solution and aims at describing the current state, and to-be state as well as a potential high-level project scope and timeline for <Service Provider>.

The document may refer to a 'Workload', which summarizes the full technical solution for a customer (You) during a single engagement. The Workload is described in the chapter Workload Requirements and Architecture.

This is a living document, additional sections will be added as the engagement progresses resulting in a final Document to be handed over to the <Service Provider>.

Business Context

Guide:

Describe the customer's business and background. What is the context of the customer's industry and LoB? What are the business needs and goals that this Workload is an enabler for? How does this technical solution impact and support the customer's business goals? Does this solution support a specific customer strategy, or maybe certain customer values? How does this solution help our customers to either generate more revenue or save costs?

Consider the different context of a new deployment of Exadata Cloud@Customer vs. an expansion of an existing deployment.

2.1 Executive Summary

Guide:

A section describing the Oracle differentiator and key values of the solution of our solution for the customer, allowing the customer to make decisions quickly.

2.2 Workload Business Value

Guide:

A clear statement of specific business value as part of the full workload scope. Try to keep it SMART: Specific, Measurable, Assignable, Realistic, and Time-Related - Agree on the business value with the customer. Keep it business-focused and speak the language of the LoB which benefits from this Workload: "Increase Customer Retention by 3% in the next year" or "Grow Customer Base with Executive Decision-Making from our Sales and Support Data". Avoid technical success criteria such as "Migrate App X to Oracle Cloud" or "Provision 5 Compute Instances". Avoid Oracle success criteria and language "Get Workload Consuming on OCI".

Workload Requirements and Architecture

3.1 Overview

Guide:

Describe the Workload: What applications and environments are part of this Workload migration or new implementation project, and what are their names? The implementation will be scoped later and can be a subset of the Solution Definition and

proposed overall solution. For example, a Workload could exist of two applications, but the implementer would only include one environment of one application. The workload chapter is about the whole Workload and the implementation scope will be described later in the chapter Solution Scope.

Database or Storage Server expansions: describe the requirements for the new workloads. Is it a new workload that will run a new VM Clusters deployed on new Database Servers/storage nodes? Is it an increasing resource demanding for an already active workload, and it will need to extend the existing vm clusters?

3.2 <Customer> Non-Functional Requirements

Guide:

Describe the high-level technical requirements for the Workload. Consider all sub-chapters, but decide and choose which Non-Functional Requirements are necessary for your engagement. You might not need to capture all requirements for all sub-chapters.

This chapter is for describing customer-specific requirements (needs), not to explain Oracle solutions or capabilities.

3.2.1 Regulations and Compliances Requirements

Guide:

This section captures specific regulatory or compliance requirements for the Workload. These may limit the types of technologies that can be used and may drive some architectural decisions.

The Oracle Cloud Infrastructure Compliance Documents service lets you view and download compliance documents:

<https://docs.oracle.com/en-us/iaas/Content/ComplianceDocuments/Concepts/compliancedocsoverview.htm>

If there are none, then please state it. Leave the second sentence as a default in the document.

Example:

At the time of this document creation, no Regulatory and Compliance requirements have been specified.

In addition to these requirements, the CIS Oracle Cloud Infrastructure Foundation Benchmark, v1.2 will be applied to the Customer tenancy.

3.2.2 Environments

Guide:

A diagram or list detailing all the required environments (e.g. development, test, live, production, etc.).

If you like to describe a current state, you can use or add the chapter 'Current State Architecture' before the 'Future State Architecture'.

Example:

There are three environments that needs to be covered in the solution.

Name	Location	Data Center	DR	High Availability	RPO	RTO
Production	Stockholm	DC1 and DC2	Yes	Yes	~0	2 hrs
Non-Production	Stockholm	DC1 or DC2	No	Yes	24 hrs	24 hrs
Miscellaneous	Stockholm	DC1 only	No	Yes	4 hrs	12 hrs

3.2.3 High Availability and Disaster Recovery Requirements

Guide:

This section captures the resilience and recovery requirements for the Workload. Note that these may be different from the current system.

The Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirement of each environment should be captured in the environments section above, and wherever possible.

- What are the RTO and RPO requirements of the Application?
- What are the SLAs of the application?
- What are the backup requirements

Note that if needed, this section may also include an overview of the proposed backup and disaster recovery proposed architectures.

Example:

Oracle MAA Best Practices and Reference Architectures should be considered for all environments.

All production databases need to be highly available and covered by a disaster recovery solution across the two data centers in Stockholm, Your Data Center 1 and Your Data Center 2. Performance needs to be maintained during failure of components or systems.

Databases in all environments should be unaffected by maintenance.

3.2.4 Backup and Recovery requirements

Guide:

This section captures the backup and recovery requirements for the Workload. Note that these may be different from the current system.

The Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirement of each environment should be captured in the environments section above, and wherever possible.

- *What is the RTO and RPO requirements for the backup and recovery?*
- *What is the backup retention requirements*

Example:

Current backup solution with NFS shares should be used initially, but the solution needs to be able to be upgraded with Zero Data Loss Recovery Appliance to accommodate increasing demands for Ransomware Protection (Compliance and Immutability) and Zero RPO.

Environment	RPO	RTO
Production	~0	2 hrs
Non-Production	24 hrs	24 hrs
Miscellaneous	4 hrs	12 hrs

3.2.5 Data Center requirements

Guide:

Capture which data centers is going to be used. And for each data center capture the current deliverables on Power delivery, Floor design and Network infrastructure.

The Exadata Cloud@Customer system needs to be configured with the correct PDU (Power Delivery Unit) to match both the power consumption of the ExaDB-C@C system as well as the delivery in each data center.

The Exadata Cloud@Customer system needs to be configured with the correct network cards and/or transceivers to match the speed of customer network and type of optical fibers in each data center.

If customer can not use Oracle racks, then re-racking is required. This requires a technical exception which needs to include OECA output and re-racking checklist.

These are items that are covered in the Exadata Cloud@Customer Network and Deployment Workshop.

Please note that 100 Gbps connection is only supported on the client and backup networks.

<CUSTOMER>	Floor type (raised or solid)	Power delivery (3-phase or 1-phase)	Rack type (Oracle or customer provided)
Data Center 1 <NICKNAME>			
Data Center 2 <NICKNAME>			

Data Center 1 <NICKNAME >	Speed (10 or 10/25 or 100 Gbps)	Cabling (optical or copper)	Optical fiber type (multimode or singlemode)
Client network			

Backup network			
Control Plane Server (CPS) network			

Example:

Current data centers (Your Data Center 1 and Your Data Center 2) must be used because of data residency requirements and application (to database) latency requirements. The databases / database services need to adhere to the data center requirements present.

Your Customer	Floor type (raised or solid)	Power delivery (3-phase or 1-phase)	Rack type (Oracle or customer provided)
Your Data Center 1	Raised	3-phase	Oracle
Your Data Center 2	Raised	3-phase	Oracle

Your Data Center 1	Speed (10 or 25 Gbps)	Cabling (optical or copper)	Optical fiber type (multimode or singlemode)
Client network	10 Gbps	Optical	Multimode
Backup network	25 Gbps	Optical	Multimode
Control Plane Server (CPS) network	10 Gbps	Optical	Multimode

Your Data Center 2	Speed (10 or 25 Gbps)	Cabling (optical or copper)	Optical fiber type (multimode or singlemode)
Client network	10 Gbps	Optical	Multimode
Backup network	25 Gbps	Optical	Multimode
Control Plane Server (CPS) network	10 Gbps	Optical	Multimode

3.2.6 Security Requirements

Guide:

Capture the Non-Functional Requirements for security-related topics. The requirements can be (but don't have to be) separated into:

- Identity and Access Management
- Data Security

Other security topics, such as network security, application security, or others can be added if needed.

Make sure to capture the requirements on Key Management as TDE (Transparent Data Encryption) is mandatory on all databases in Exadata Cloud@Customer.

ExaDB-C@C needs to connect to OCI Region so capture security requirements for this connection. For example, if Fast Connect or Site-to-Site VPN is required. For a new deployment this would need to be configured prior to Data Center Site Inspection and IP Checks.

Example:

Site-to-Site encryption required, security policies do not allow for public internet connections. This can either be Site-to-Site VPN or preferably a dedicated and private connection.

The existing installation of Oracle Key Vault (OKV) must be used.

3.2.7 Operator Access Control

Guide:

This is an optional section.

Make sure to capture the requirements on Operator Access Control, as described here:

[Oracle Operator Access Control for Exadata Cloud@Customer](#)

Example:

All operations from the outside to the system needs to be under strict control by customer in case of crisis or detection of malicious attempts. Restricted access and monitoring of all operations need to be controlled by the customer.

3.2.8 Networking Requirements

Guide

Capture the Non-Functional Requirements for networking-related topics. You can use the networking questions in the Annex. As businesses increasingly rely on Cloud Infrastructure to store, process, and transmit sensitive data, the need for comprehensive security solutions has never been more important. Potential customers evaluating network security solutions typically prioritize the following requirements: Some of the broader category considerations are below.

- *Data Protection: Safeguarding sensitive information against unauthorized access, theft, or modification is a primary concern for any organization and industry today.*
 - *Threat Prevention: Advanced capabilities like IDPS and malware detection for blocking threats.*
 - *Data Loss Prevention (DLP): Monitoring and controlling sensitive data transmission.*
 - *Encryption and Decryption: Inspecting encrypted traffic without compromising privacy.*
- *Threat Prevention: Proactively identifying and mitigating security threats is essential for maintaining the integrity of network infrastructure.*
 - *Intrusion Detection and Prevention: Monitoring for suspicious or malicious activity.*
 - *Application Control: Granular control over specific applications or services.*
 - *URL Filtering: Controlling access to permitted URLs.*
- *Security compliance: Does your organization have network security requirements based on industry or organization compliance? For example - SAMA (Saudi Arabia Monetary Authority), HIPAA (Health Insurance Portability and Accountability Act), GDPR (General Data Protection Regulation), SWIFT, etc.*

Try to capture network infrastructure requirements in the Data Center requirements section above.

Example:

Site-to-Site encryption is required, security policies do not allow for public internet connections. This can either be a private Site-to-Site VPN or preferably a dedicated connection like Fast Connect.

3.2.9 Management and Monitoring

Guide:

This subsection helps you capture any requirements for customer management and monitoring needs - e.g. system monitoring, systems management, log analysis, etc.

When you move or start an OCI project, you have a choice to use the tools you are familiar with (should they support modern application architectures), replace them with OCI native Observability services, or use a combination to improve your visibility. When contemplating how to proceed, here are some general questions that will guide you:

- *Does the tool manage across hybrid and multi-cloud environments?*
- *What is the cost of integrating the existing tool with OCI?*
- *Is my current monitor tool enabling you to prevent issues versus reacting to them?*
- *Does the tool tell you how much impact there has been on users or just that there was an impact like something is down or unavailable?*
- *Does the tool provide the full vision of applications and their infrastructure or just a piece of them or specific technology?*

Example:

Task	Target	Location	New	Notes
Application Monitoring	All targets	On-Prem and OCI	No	
Monitoring	All targets	OCI (Migration)	No	
Log Management	All targets	OCI (Migration)	No	
Insight	All Oracle DBs	OCI (Migration)	No	

3.3 Future State Architecture

Guide:

The Workload Future State Architecture can be described in various forms. In the easiest case, we describe a Logical Architecture, possibly with a System Context Diagram. A high-level physical architecture is mandatory as a description of your solution.

This should be the final architecture as part of the pre-sales solution, not an intermediate or draft version

Additional architecture, in the subsections, can be used to describe needs for specific workloads.

3.3.1 Mandatory Security Best Practices

Guide:

Use this text for every engagement. Do not change. Aligned with the Cloud Adoption Framework

The safety of <Customer Name>'s Oracle Cloud Infrastructure (OCI) environment and data is the <Customer Name>'s priority. The following table of OCI Security Best Practices lists the recommended topics to provide a secure foundation for every OCI implementation. It applies to new and existing tenancies and should be implemented before the Workload defined in this document will be implemented.

Workload-related security requirements and settings like tenancy structure, groups, and permissions are defined in the respective chapters.

Any deviations from these recommendations needed for the scope of this document will be documented in the chapters below. They must be approved by <Customer Name>.

<Customer Name> is responsible for implementing, managing, and maintaining all listed topics.

CATEGORY	TOPIC	DETAILS
User Management	IAM Default Domain	Multi-factor Authentication (MFA) should be enabled and enforced for every non-federated OCI user account. - For configuration details see Managing Multi-Factor Authentication. In addition to enforcing MFA for local users, Adaptive Security will be enabled to track the Risk Score of each user of the Default Domain. - For configuration details see Managing Adaptive Security and Risk Providers.
	OCI Emergency Users	A maximum of three non-federated OCI user accounts should be present with the following requirements: - Username does not match any username in the Customer's Enterprise Identity Management System - Are real humans. - Have a recovery email address that differs from the primary email address. - User capabilities have Local Password enabled only. - Has MFA enabled and enforced (see IAM Default Domain).
	OCI Administrators	Daily business OCI Administrators are managed by the Customer's Enterprise Identity Management System. This system is federated with the IAM Default Domain following these configuration steps: Federation Setup

CATEGORY	TOPIC	DETAILS
Cloud Posture Management		- User Provisioning - For configuration guidance for major Identity Providers see the OCI IAM Identity Domain tutorials.
	Application Users	Application users like OS users, Database users, or PaaS users are not managed in the IAM Default Domain but either directly or in dedicated identity domains. These identity domains and users are covered in the Workload design. For additional information see Design Guidance for IAM Security Structure .
	OCI Cloud Guard	OCI Cloud Guard will be enabled at the root compartment of the tenancy home region. This way it covers all future extensions, like new regions or new compartments, of your tenancy automatically. It will use the Oracle Managed Detector and Responder recipes at the beginning and can be customized by the Customer to fulfill the Customer's security requirements. - For configuration details see Getting Started with Cloud Guard. Customization of the Cloud Guard Detector and Responder recipes to fit the Customer's requirements is highly recommended. This step requires thorough planning and decisions to make. - For configuration details see Customizing Cloud Guard Configuration
	OCI Vulnerability Scanning Service	In addition to OCI Cloud Guard, the OCI Vulnerability Scanning Service will be enabled at the root compartment in the home region. This service provides vulnerability scanning of all Compute instances once they are created. For configuration details see Vulnerability Scanning.
Monitoring	SIEM Integration	Continuous monitoring of OCI resources is key for maintaining the required security level (see Regulations and Compliance for specific requirements). See Design Guidance for SIEM Integration to implement integration with the existing SIEM system.
Additional Services	Budget Control	OCI Budget Control provides an easy-to-use and quick notification on changes in the tenancy's budget consumption. It will be configured to quickly identify unexpected usage of the tenancy. For configuration details see Managing Budgets

3.3.2 Naming Conventions

Guide:

This chapter describes naming convention best practices and usually does not require any changes. If changes are required please refer to [Landing Zone GitHub](#). The naming convention zone needs to be described in the Solution Design by the service provider.

Use this chapter ONLY for new cloud deployments and remove it for brownfield deployments.

Naming convention is an important part of any deployment to ensure consistency, governance, and security within your tenancy. Find [here](#) Oracle's recommended best practices.

3.3.3 OCI Landing Zone Solution Definition

Guide:

This chapter describes landing zone best practices and usually does not require any changes. If changes are required please refer to [Landing Zone GitHub](#). The full landing zone needs to be described in the Solution Design by the service provider. Use this template ONLY for new cloud deployments and remove it for brownfield deployments.

Database or Storage Server expansions: if it doesn't apply to the expansion activities, mark the sections as "N/A due to limited-scope activity: Only Expansions Needed."

To reduce your time-to-production to deploy ExaDB-C@C workloads, you can get guidance and fulfil IAM and Observability requirements by checking the [ExaDB-C@C Landing Zone Workload Extension](#). For getting the latest information you can check the [GitHub repository](#).

Example:

An OCI Landing Zone sets the foundations for a secure tenancy, providing design best practices and operational control over OCI resources. A Landing Zone also simplifies the onboarding of workloads and teams, with clear patterns for network isolation and segregation of duties in the organization, which sets the cloud operating model for day-to-day operations. Oracle highly recommends the use of an OCI Landing Zone for any deployment. Use these guidelines to set up your OCI Landing Zone, including design considerations, approaches, and solutions to use.

Note that all workloads in a tenancy should sit on top of a Landing Zone, meaning that the workload architecture defined in the next section can be subject to adjustments (e.g., network structure) towards the landing zone model, along with other future workloads.

3.3.4 Logical Solution Architecture

Guide:

Provide a high-level logical Oracle solution for the complete Workload. Indicate Oracle products as abstract groups, and not as physical detailed instances. Create an architecture diagram following the latest notation and describe the solution.

To implement a solution the Physical Architecture is needed in the next chapter. The physical notation can show individual components with physical attributes such as IP addresses, hostnames, or sizes.

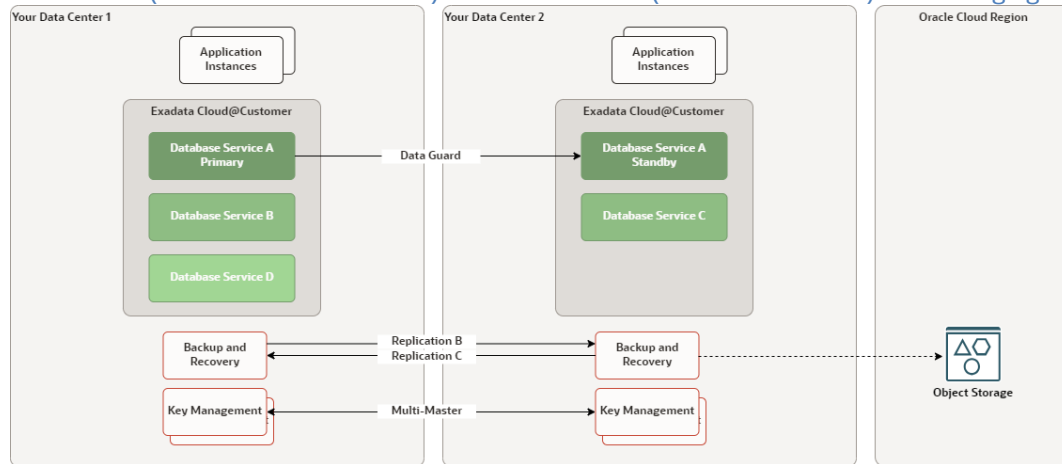
The Oracle Cloud Notation, OCI Architecture Diagram Toolkits

Example:

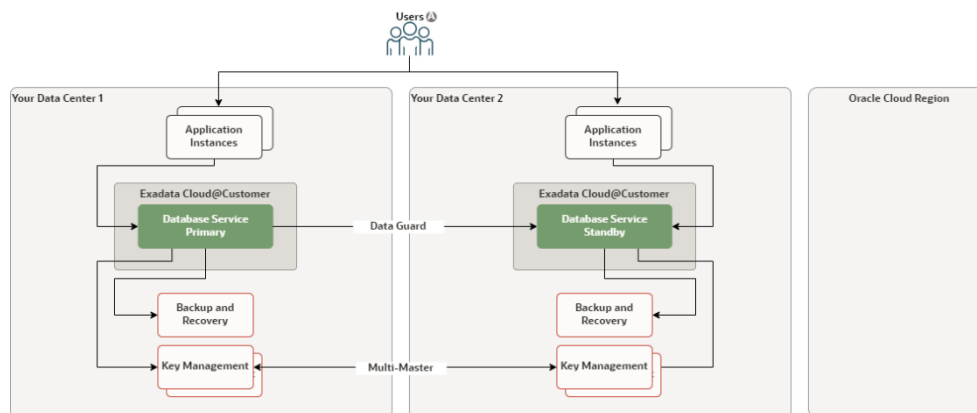
Database consolidation with multiple service levels.

Based on requirements for data residency and local application performance this solution delivers Database as a Service (DBaaS) in Your Data Centers using Exadata Cloud@Customer. Considering Oracle Maximum Availability Architecture (MAA) multiple service levels have been defined in the logical solution architecture below.

The logical architecture have multiple service levels for Production (Database Service A), non-Production divided between data centers (Database Service B and C) and Miscellaneous (Database Service D) due to ranging requirements.



3.3.4.1 Service Level with High Availability, Disaster Recovery and Data Protection



Details of Database Service A based on MAA Reference Architecture "Gold".

3.3.5 Physical Architecture

Guide:

The Workload Architecture is typically described in a physical form. This should include all solution components. You do not have to provide solution build or deployment details such as IP addresses.

Please describe the solution with an architecture image plus a written text. If you have certain specifics you like to explain, you can also use the Solution Consideration chapter to describe the details there.

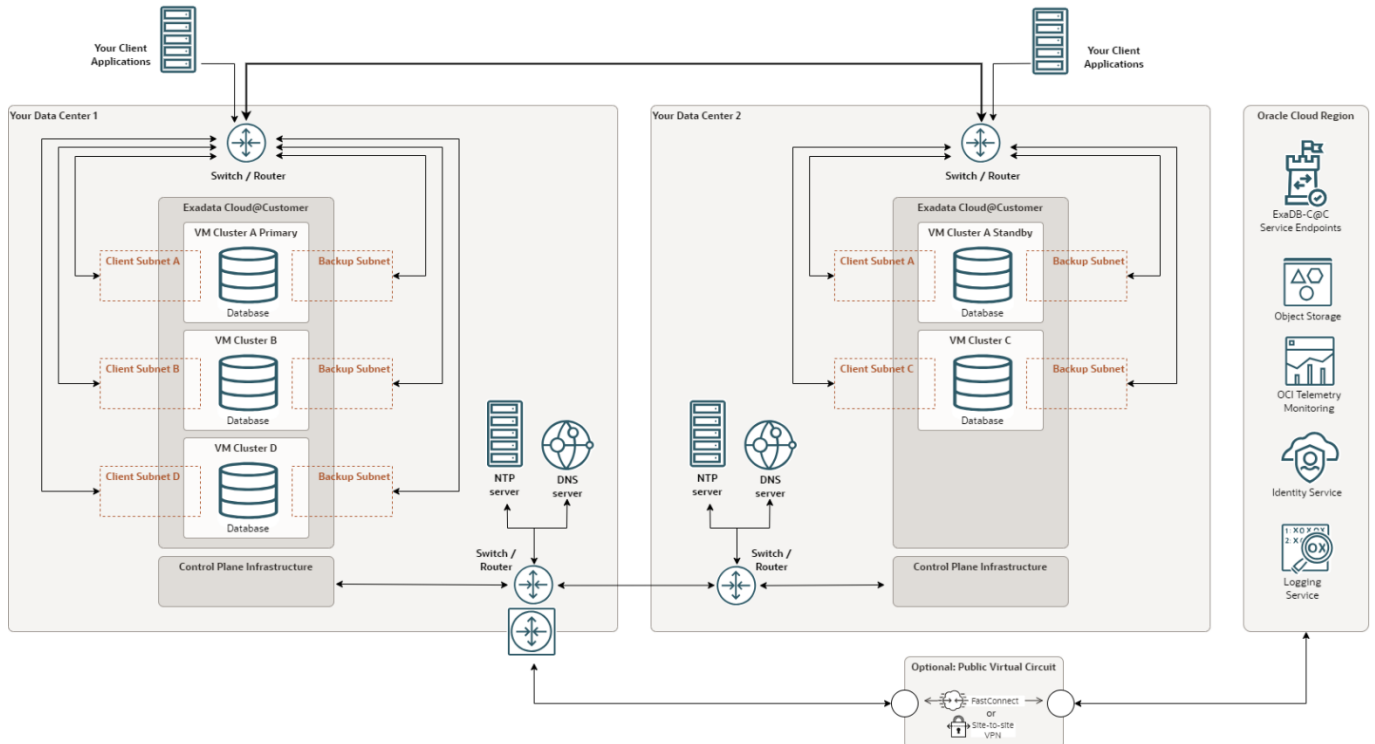
[The Oracle Cloud Notation, OCI Architecture Diagram Toolkits](#)

Reference: [StarterPacks \(use the search\)](#)

Example:

Database consolidation with multiple service levels.

Based on the defined logical architecture the physical architecture to implement consists of two Exadata Cloud@Customer systems, one in each of Your Data Centers. To implement the required MAA Reference Architecture service levels there will be three VM Clusters in ExaDB-C@C in Your Data Center 1 and two VM Clusters will be created in ExaDB-C@C in Your Data Center 2.



The VM Clusters fulfill the following requirements:

- Network segmentation due to security.
Each VM Cluster have its own Client Subnet including VLAN ID.
- Workload isolation
Each VM Cluster have its own dedicated resources in CPU, Memory and Storage Capacity.
- Application / User separation
Each VM Cluster can have its own users and groups as defined in the Identity Service.
- Required Service Level Objectives (for example MAA Gold, Silver)
Each VM Cluster is designed against a given MAA Reference Architecture.

3.3.6 Key Solution Proposal Components

Our proposal is based on the following main technical components:

3.3.6.1 Exadata Cloud@Customer (ExaDB-C@C)

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:

Oracle Exadata Cloud@Customer brings the performance, automation, and economics of Exadata Database Service and Autonomous Database fully managed by Oracle to data centers in <CUSTOMER NAME>. This is the easiest way for <CUSTOMER NAME> to start using cloud database resources in their data centers while meeting strict data residency standards. Exadata Cloud@Customer incorporates unique optimizations that enable databases to run faster with less management and at lower costs, to better leverage <CUSTOMER NAME> data.

Value Proposition

Oracle's value proposition to support <CUSTOMER NAME>'s needs is built around the Oracle Exadata Cloud@Customer cloud platform:

Resilience and Availability

- Modern, highly available, and secure data architecture by design
- Services installed in <CUSTOMER NAME> data center, dedicated and isolated, to meet data sovereignty requirements
- Simplified patch maintenance and proactive threat management

Performance, agility, and flexibility

- Low latency to keep data as close as possible to applications and choose the timeline for modernizing these databases as needed
- Ability to scale the platform as needs change without increasing complexity
- Maximum availability architecture (e.g. PRA with fast RTO and RPO, service continuity, etc.)
- Obsolescence and risks
- Simplify and standardize to enhance security
- Automation to achieve more predictable service levels and address a wide range of security vulnerabilities
- Encryption of all data at rest and in transit, with BP2i control of encryption keys.
- Ability to use an existing HSM, ensuring that <CUSTOMER NAME> has data privacy, and Oracle Key Vault between that HSM and the databases, to have the full level of integration offered by the provisioning and management tools of the Exadata Cloud@Customer architecture

Note: OKV is fully integrated with the Exadata Cloud@Customer solution and enables the use of all management APIs. Because these are maintained by Oracle, this causes a decrease in TCO. By using OKV, Oracle provides support and onboarding only, without the need for a third-party contact. From a long-term perspective, OKV is the only possible solution to date for outsourced key management for Autonomous Database

Speed of implementation / Time to Market

- Time to Market optimization is made possible thanks to the comprehensive "Infrastructure as Code" approach, which allows all environments to be managed through a DevOps methodology to ensure repeatability.
- Implementation of applications directly and without the need for infrastructure teams, greatly accelerating their implementation and simplifying their maintenance.

Access to innovative services and a unified data platform

Innovation around data services has been at the heart of Oracle's DNA since its inception. Beyond access to all

cloud services around data, <CUSTOMER NAME> will be able to benefit from innovation around the Oracle database, which coupled with cloud services and innovation allows for implementation in its "Autonomous" version. Beyond modernization, some use cases are potentially better served by the direct use of the PaaS service available on the Exadata Cloud@Customer platform, with a project time potentially shorter or equivalent to reproducing the existing or not benefiting from the PaaS services

Note: In the context of <CUSTOMER NAME>, this is a parallel study to the implementation to benefit from innovation capabilities, through the use of Autonomous Database instead of other database services where possible, to bring short-term benefits without undue complexity to migration projects.

- Align with Sustainable Development Goals
- Aware of climate issues, Oracle has set targets to reduce CO2 production, both internally at the group level and for all of its partners.

Oracle is committed to:

- Achieving carbon neutrality in 2025
- Reduce CO2 production by 50% in 2030
- Reaching Net Zero in 2050
- Efficiency of Hardware Oracle OCI ExaDB-C@C solutions: 4d eco-design program, certifications, etc.
- Renewable energy use for all global operations by 2025
- End-to-end infrastructure reuse by our engineers: 4d eco-design program, supply chain redesign for delivery and recovery with 99.9% reuse/recycling rate in 2022
- Reduction in the average measured carbon footprint (CSR audit firm) of our customers migrating to our cloud solutions in Europe: 40%

3.3.6.2 Backup and Recovery Options

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:

The Logical Architecture describes the high-level backup and recovery solution.

Oracle Exadata Cloud@Customer offers two approaches to configure and take backups as a recommended solution: Oracle Managed Backup and User Configured Backup.

- Oracle Managed Backup: Database backups are managed entirely by the Oracle Exadata Cloud@Customer service based on a one-time configuration.
- User Configured Backup: It is the user's responsibility to configure and execute backup operations according to their preferences.

Customers can also choose from different Backup Destinations

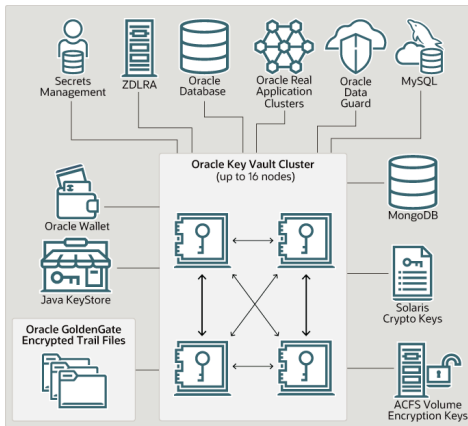
- Customer Managed Zero Data Loss Recovery Appliance backup destination
- Customer Managed NFS backup destination
- Oracle Cloud Infrastructure object storage
- Local Exadata storage on your Exadata Database Service on Cloud@Customer system

3.3.6.3 Oracle Key Vault

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:



Oracle Key Vault is engineered to deliver performant, fault-tolerant, and flexible encryption key management for transparent data encryption (TDE) as part of Oracle Advanced Security. Key Vault works with all database deployment options, including Oracle Real Application Clusters (Oracle RAC), Oracle Data Guard, sharded databases, RMAN backups, and Oracle Multitenant pluggable databases. Key Vault scales to support highly consolidated workloads on engineered systems, such as Oracle Exadata, Exadata Database Service on Dedicated Infrastructure (ExaDB-D) as well as full integration into the database provisioning workflow for Exadata Cloud@Customer (ExaDB-C@C) and Autonomous Database on Exadata Cloud@Customer (ADB-C@C).

We can therefore provide <CUSTOMER NAME> with assurance that Oracle Databases will be deployed on the best environment to ensure the availability and sustainability of critical data and to meet European requirements for regulating and resuming banking activities in the event of cyber-attacks.

3.4 Solution Considerations

Guide:

Describe certain aspects of your solution in detail. What are the security, resilience, networking, and operations decisions you have taken that are important for your customer?

3.4.1 High Availability and Disaster Recovery

Guide:

- Resilience on OCI.
- Workload Related Content.
- What happens when site 1 is down?
- What happens when XYZ?
- How is HA achieved?

Example:

Customer requires to adhere to Oracle MAA principles.

Following has been decided:

- Production Databases:
 - Database Service A
 - MAA Gold with 2-node dedicated VM-cluster in DC1 and 2-node dedicated VM-Cluster in DC2
 - Full RAC
 - Data Guard with Your DC1 as primary DC and Your DC2 as standby DC
 - Backups will be made from the local instance to the local backup environment.
- Non-Production Databases:

- Database Service B & C
- MAA Silver with 2-node VM-Cluster in Your DC1 for Database Service B and 2-node VM-Cluster in Your DC2 for Database Service C
- Database Service B and Database Service C spread across DC for efficiency.
- Full RAC
- Local backups that will be replicated through the backup solution to your other DC.

3.4.2 Backup and Recovery

Guide:

Exadata Cloud@Customer requires a backup network, with the intent to run Automatic or Manual backups of databases. Describe the intended backup solution.

Example:

Customer has standardized backups for Oracle DB based on custom RMAN scripts sending DB backups to external NFS share. In the first phase DB backups from the ExaDB-C@C need to be integrated with and stored on the external NFS share(s). Customer decided to work with Oracle Managed Backups on the ExaDB-C@C instead of his own custom scripts. Customer requirement is to have 60-day retention for Database Service A (Production) and 30-day retention for Database Service B&C (Non-Production). Weekly full backup and daily incremental backups will be taken, full backups on Sunday, incremental backups the other days of the week. Every 30 min backups of the archive logs will be taken as well. Backup retention will be configured per customer's requirement. Backups will be stored on the external NFS share(s) that the customer will continue to manage. In a later phase, to meet the RPO close to zero requirement, Oracle will work with the customer to evaluate and build a case around Oracle Zero Data Loss Recovery Appliance.

3.4.3 Data Center prerequisites

Guide:

Describe for each data center the decisions taken and how the solution have been configured to meet requirements.

What PDU type have been selected, and how many power receptacles will the customer require?

For each network, which transceivers have been selected, their speed and type, and how many ports and cables will the customer require?

For the proposed ExaDB-C@C hardware shape, document the power and cooling requirements (for example, use OECA output). Link to OECA: <https://www.oracle.com/database/technologies/oeca-download.html>

Floor space – Make sure there is adequate floor space for service clearance.

Documented here: [Site Requirements for Oracle Exadata Database Service on Cloud@Customer](#)

Copy the selected table in the Excel Spreadsheet and paste. As Image or as a new table.

And screenshot the OECA configuration and paste as Image.

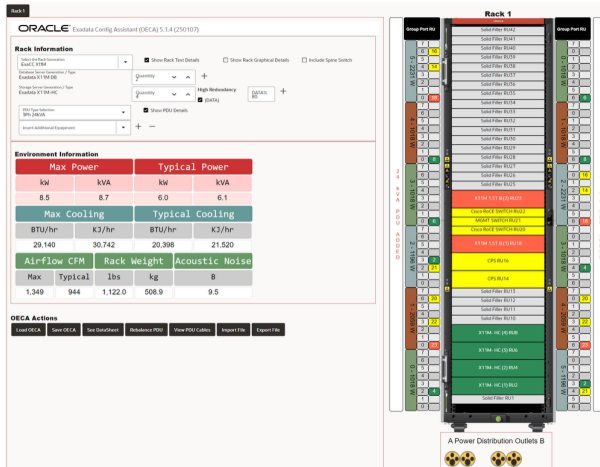
Example:

The table below summarizes the connectivity that will be required for each datacenter.

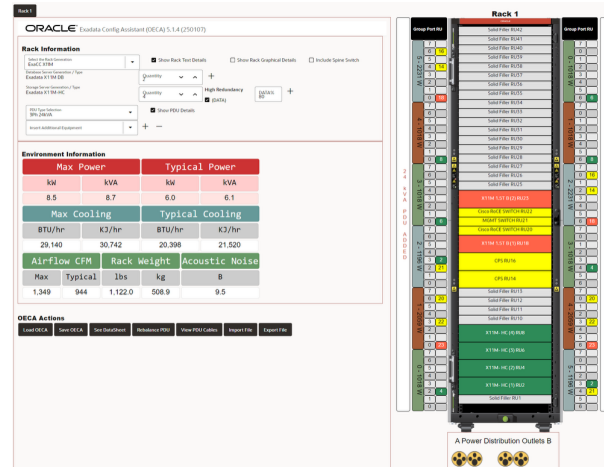
			Power		Network ports					To be provided by CUSTOMER			
YOUR DC1	Rack	Racks	3 Phase HV 24 kVA PDU	Whips	Client Access/DataGuard 10G	Backup 25G	DataGuard	CPS 10G	Total	Power Receptacles	Optical Cables LC-LC	Transceivers 10GbE SFP+	Transceivers 25GbE SFP28
	EXACC1	1	2	4	4	4	0	4	12	4	12	8	4
	TOTAL	1	2	4	4	4	0	4	12	4	12	8	4

			Power		Network ports					To be provided by CUSTOMER			
YOUR DC2	Rack	Racks	3 Phase HV 24 kVA PDU	Whips	Client Access/DataGuard 10G	Backup 25G	DataGuard	CPS 10G	Total	Power Receptacles	Optical Cables LC-LC	Transceivers 10GbE SFP+	Transceivers 25GbE SFP28
	EXACC2	1	2	4	4	4	0	4	12	4	12	8	4
	TOTAL	1	2	4	4	4	0	4	12	4	12	8	4

Customer also ensured that the datacenter facilities can deliver the required power and cooling to host the ExaDB-C@C racks. Oracle Exadata Config Assistant (OECA) has been used to detail the specific requirements for both datacenters.



Your Datacenter 1



Your Datacenter 2

Oracle documentation link: [Preparing for Exadata Database Service on Cloud@Customer](#)

- Oracle Cloud Infrastructure (OCI) Requirements for Oracle Exadata Database Service on Cloud@Customer
Learn the basic concepts to get started using Oracle Cloud Infrastructure.
- Site Requirements for Oracle Exadata Database Service on Cloud@Customer
Review the requirements for provisioning Oracle Exadata Database Service on Cloud@Customer at your site.
- Network Requirements for Oracle Exadata Database Service on Cloud@Customer
Review the network requirements for provisioning Oracle Exadata Database Service on Cloud@Customer at your site.
- Storage Configuration Requirements for Oracle Exadata Database Service on Cloud@Customer
Review the storage requirements for the ASM disk groups or Exascale and VM file systems to plan the best options for your enterprise needs.

Checklists for Exadata Database Service on Cloud@Customer Deployments

To determine your readiness for an Exadata Database Service on Cloud@Customer deployment, review the deployment checklists.

3.4.4 Security

Please see our security guidelines in the [Annex](#).

3.4.5 Operator Access Control

Guide:

Make sure to describe if the solution requires Operator Access Control, as described here, in order to meet customer requirements.

[Oracle Operator Access Control for Exadata Cloud@Customer](#)

If not required, see example text below.

Example:

This is not applicable for this solution.

3.4.6 Networking

Guide:

A list of possible Oracle solutions can be found in the Annex.

The example text below is quite generic and can be kept as is.

Please consider changes as you read through the text.

Example:

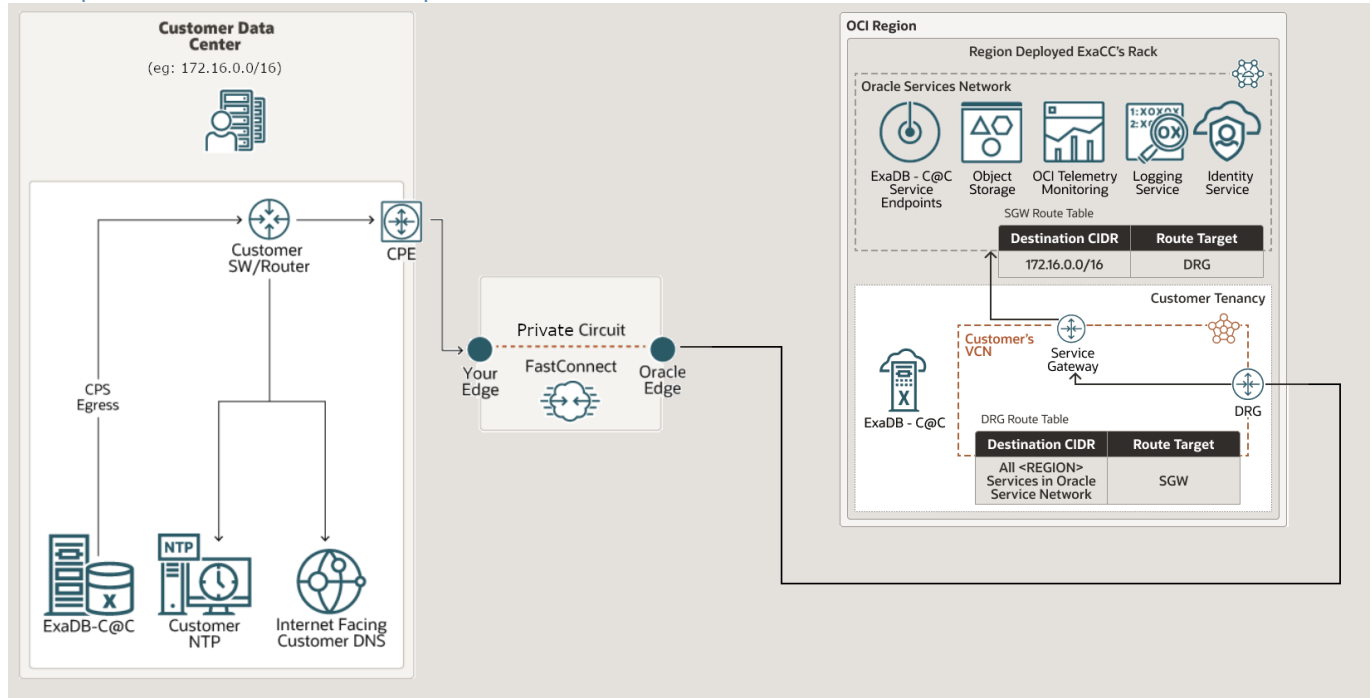
In Datacenter Networking

All networking within the customer's datacenter is optical multimode and the target Exadata Cloud@Customer will connect to the core network. Appropriate transceivers (SFP+/SFP28 modules) for the network ports on the ExaDB-C@C will be part of delivery.

Transceivers for customer network ports, patch panels and optical network fibers will be delivered by customer.

Network connectivity to OCI

Customer does not allow a connection over public internet to OCI. Therefore, it has been decided to deploy a FastConnect over a private circuit as detailed in the picture below.



Oracle Exadata Database Service on Cloud@Customer FastConnect connectivity to OCI through private peering

3.4.7 Manageability and Observability

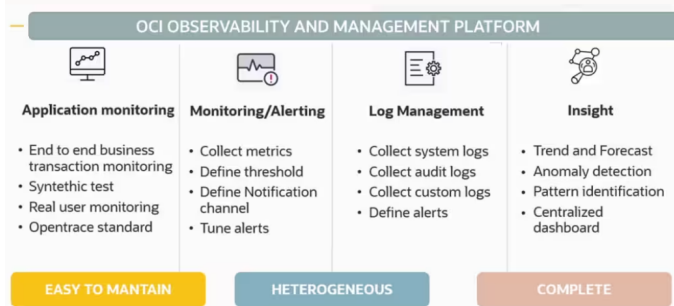
Guide:

This chapter and subchapters are generic and should be kept as is, please change the <CUSTOMER NAME>

Observability is a technology advancement focused on getting insights from a vast array of data, logs, and events generated within an IT environment. By implementing an Observability strategy, organizations gain the capability to anticipate system disruptions, prevent resource overconsumption, and enhance the overall application user satisfaction. That means being proactive, which is a must, especially in a distributed environment.

Gone are the days when the IT landscape remained a mysterious black box. The company's digitalization and the Cloud model compel C-level executives to gain comprehensive insights into asset utilization. The efficient allocation of resources directly influences budgetary considerations.

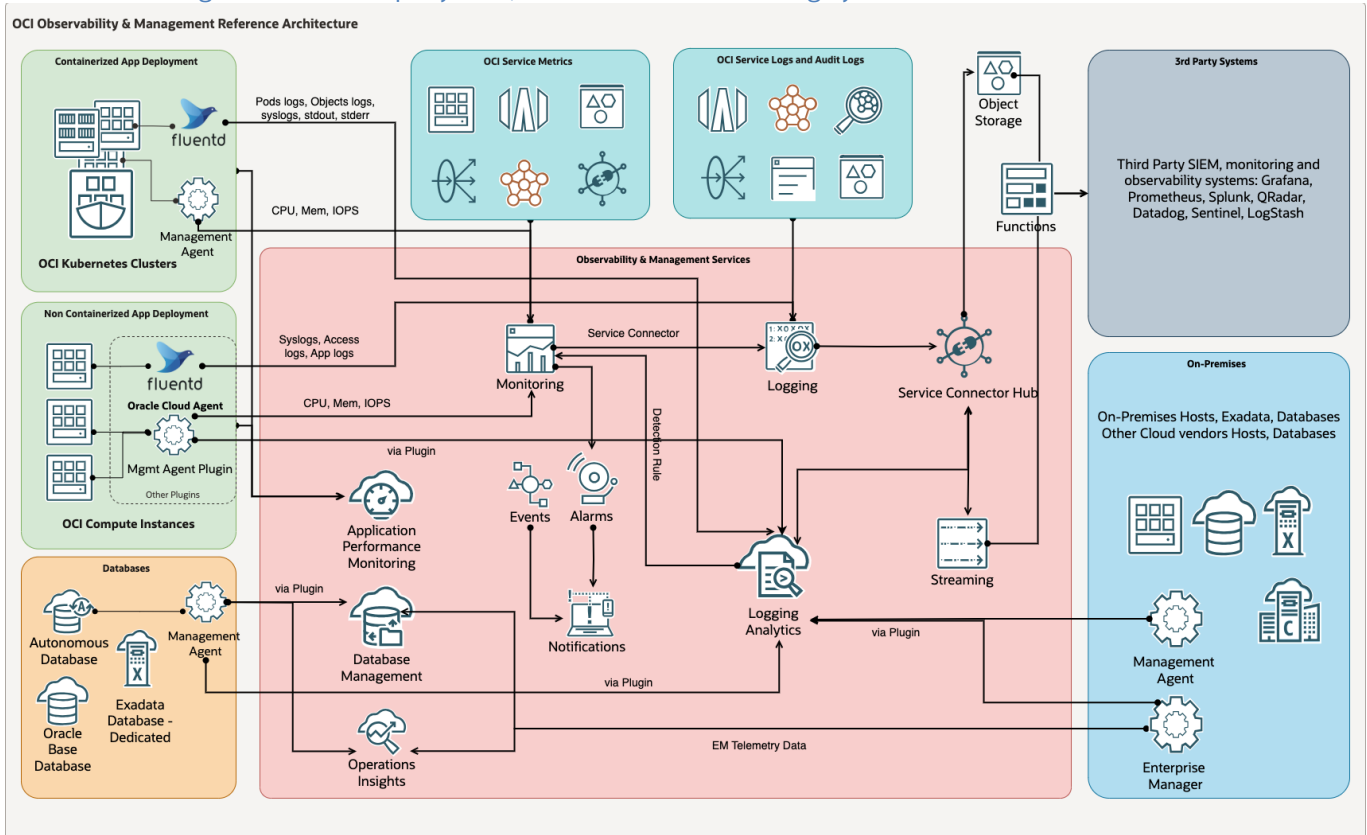
Observability helps organizations examine how well their infrastructure is working, predict future needs, and help take proactive steps to improve efficiency and protect investments. Therefore, Observability tools are needed to cover these important areas.



Observability and Manageability

3.4.7.1 Observability Architecture

The basic monitoring OCI services collect the data and send logs and metrics to OCI Monitoring and Logging services. If you want to apply machine-learning capabilities and perform analysis, you can send the data to the Logging Analytics service. If you want to use OCI Logging Analytics to collect logs coming from both on-premises and cloud sources to analyze them for auditing, security purposes, or to integrate data with an external SIEM solution, the Connector Hub serves as the solution. It's advisable to plan your monitoring strategy by considering both the O&M (Observability and Management) native service of OCI and its integration with third-party tools, as O&M is flexible and a highly customizable solution.



OCI Architecture

3.4.7.2 Real-Time Monitoring

Real-time monitoring is the delivery of continuously updated data about systems, processes, or events. Such monitoring provides information streaming at zero or low latency, so there is minimal delay between data collection and analysis. It enables quick detection of anomalies, performance issues, and critical events.

Please find all references for this chapter in the Annex.

3.4.7.3 Performance and Tuning

Performance tuning is the improvement of system performance. It can be done proactively to prevent issues or reactively in response to increased workload, which is crucial for avoiding system outages.

Please find all references for this chapter in the Annex.

3.4.7.4 Administration

Administrator tasks involve upholding a data management policy and ensuring essential equipment functionality, such as instance management, backup & restore operations, key management, and allocating resources from the database to the storage.

Please find all references for this chapter in the Annex.

3.4.7.5 Troubleshooting

Issues can happen on several levels. To identify the root cause, it is important to be able to correlate resources, drill down into the issues, and analyze trends in the systems. It's crucial to consider that the application itself might be the root cause of the issue. Therefore, it's essential to gather information about the application's behavior and performance to fully understand the problem and resolve it effectively. Troubleshooting also allows you to avoid an outage which is why it is important to notice issues as early as possible.

Please find all references for this chapter in the Annex.

3.4.7.6 Cost Control and Chargeback

Cost control is the practice of identifying and reducing business expenses to increase profits. It starts with the budgeting process. Cost control is an important factor in maintaining and growing profitability.

IT chargeback can provide greater visibility into the costs of IT services and infrastructure usage. It enables organizations to identify opportunities for cost optimization and reduce wasteful spending.

Cost control and chargeback are critical concerns, especially for companies transitioning to the cloud, presenting new financial operational challenges (FinOps). In this context, reducing consumption directly impacts the company's business.

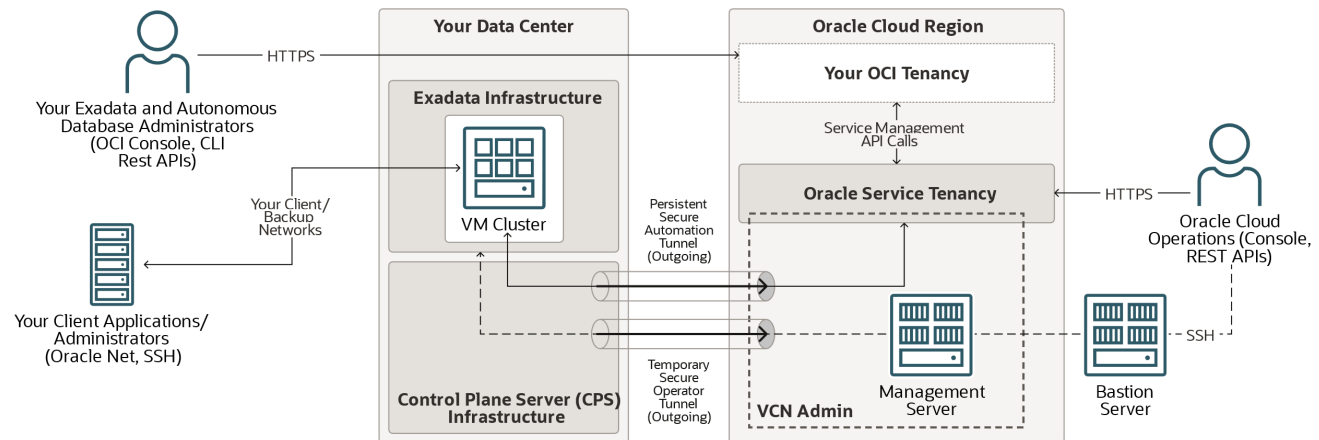
Please find all references for this chapter in the Annex.

3.5 Exadata Cloud@Customer Architecture

Guide:

This chapter and subchapters is generic and should be kept as is, please change the <CUSTOMER NAME>

3.5.1 Exadata Cloud@Customer Overview



Oracle Exadata Cloud@Customer runs Oracle Exadata Database Service and Oracle Autonomous Database Service with Oracle-owned and managed infrastructure located in your data center.

Your databases run on one or more virtual machine (VM) clusters that reside on Exadata infrastructure in your data center. Exadata Cloud@Customer also includes control plane server (CPS) infrastructure that connects to an OCI region for cloud automation and administration.

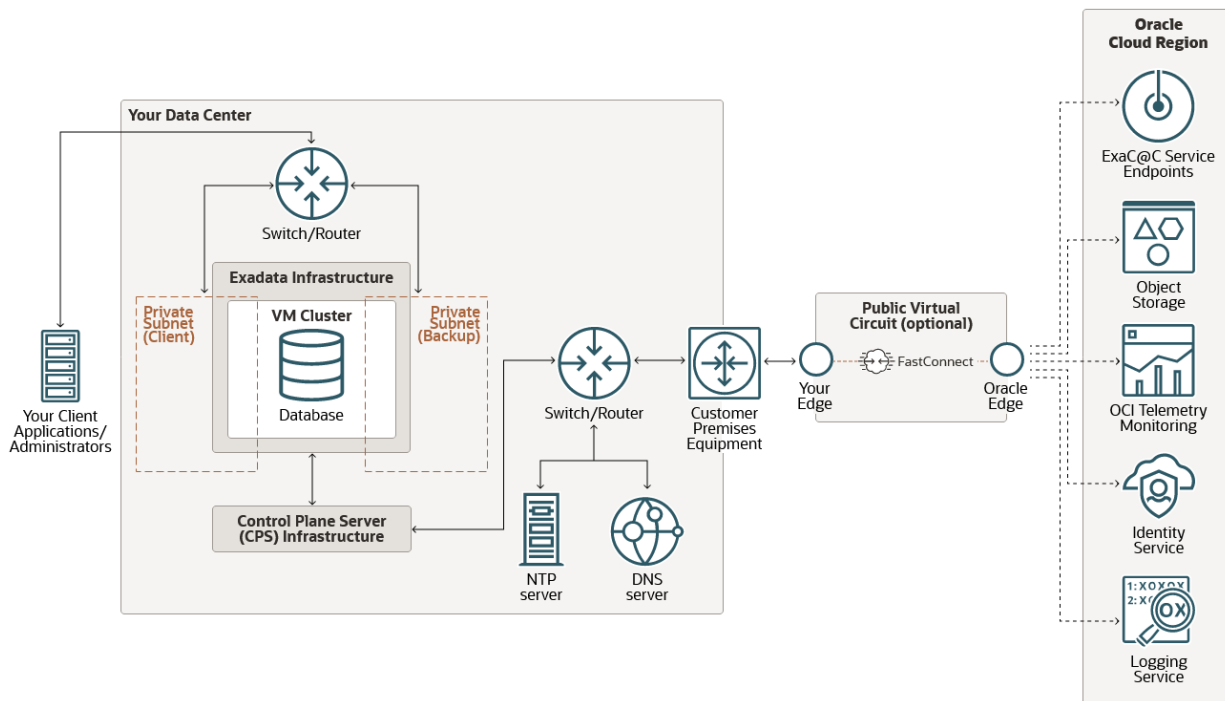
Your client applications and administrators connect to the VM clusters through client and backup networks that you create. You access your databases through standard Oracle database connection methods, such as Oracle Net. You access the VM clusters through standard Oracle Linux methods, such as token-based Secure Shell (SSH).

Your Exadata and Autonomous Database administrators can use the web-based OCI Console, command-line interface (CLI), and REST APIs to connect to your OCI tenancy over an HTTPS connection. Your tenancy sends service management API calls to the Oracle service tenancy. The Oracle service tenancy connects to the Oracle-managed admin virtual cloud network (VCN). An outgoing, persistent, secure automation tunnel connects the CPS infrastructure in your data center to the Oracle-managed admin VCN in the OCI region for delivering cloud automation commands to the VM clusters.

For Autonomous Database, separate persistent automation and temporary operator tunnels are created, targeting separate service tenancy administrative VCNs. If you're using both Oracle Exadata Database Service and Oracle Autonomous Database Service, you'll have two persistent tunnels and, as required, up to two temporary tunnels. These separate tunnels enforce a strict separation of duties for developer operations personnel, who only have authorization to take action on specific VM cluster resources, segregated by service type.

Oracle Cloud Operations can use a console or REST APIs over an HTTPS or SSH connection to manage the infrastructure. Oracle operator SSH connections travel through bastion and management servers, the admin VCN, and an outgoing, temporary, secure operator tunnel from the CPS to the admin VCN. Oracle operator HTTPS connections travel to the infrastructure through the Oracle service tenancy, admin VCN, and the outgoing, persistent, secure automation tunnel. Oracle operator SSH connections travel through bastion and management servers; the admin VCN; and an outgoing, temporary, secure operator tunnel from the CPS to the admin VCN. You can optionally control authorization of SSH access by Oracle operators by registering infrastructure resources with a separate Operator Access Control.

3.5.2 Exadata Cloud@Customer Network



Your databases run on one or more virtual machine (VM) clusters that reside on Exadata infrastructure in your data center. Exadata Cloud@Customer also includes control plane server (CPS) infrastructure that connects to an Oracle Cloud Infrastructure (OCI) region for cloud automation and administration.

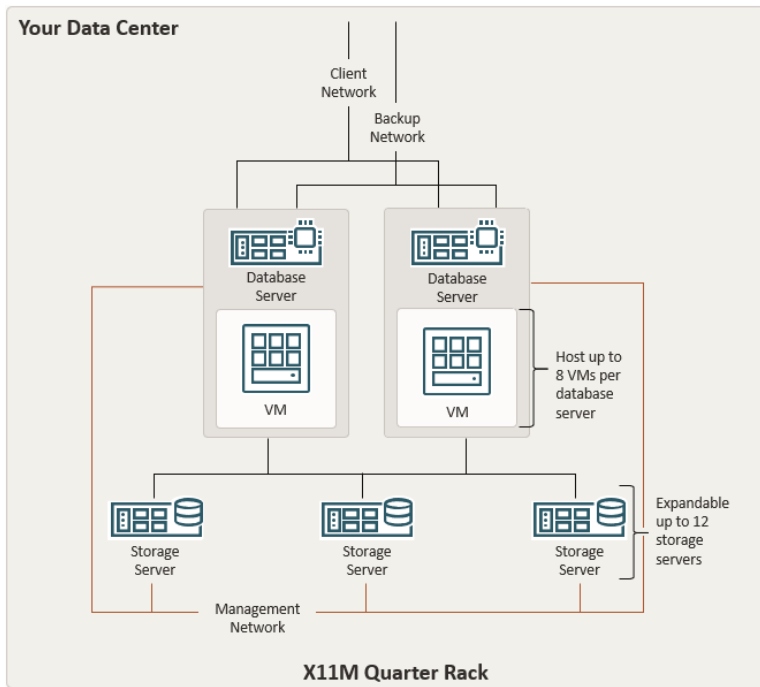
Your data center connects to the OCI region to access Exadata Cloud@Customer (ExaC@C) service endpoints, Object Storage, OCI telemetry monitoring, identity service, and logging service.

You can optionally connect your data center to the OCI region with FastConnect. This option uses a public virtual circuit to connect the edge of your network to the Oracle edge in the OCI region. Your edge connects to customer-premises equipment (CPE) in your data center.

The CPE connects to a switch or router, which directs traffic to and from the CPS, network time protocol (NTP) server, domain name system (DNS) server. If you're not using FastConnect, your network should also include a proxy server (not shown in the diagram).

Your client applications and administrators connect to your VM clusters and databases over the private client and backup subnets that you access through a switch or router in your data center. You create and manage the client and backup subnets.

3.5.3 Exadata Cloud@Customer Rack Overview



Oracle Exadata Cloud@Customer includes multiple database servers and Exadata storage servers connected by high-speed, low-latency network fabric. The Exadata rack resides in your data center.

You can choose from two elastic system shapes: elastic base and elastic. Each system shape can be scaled with additional database and storage servers.

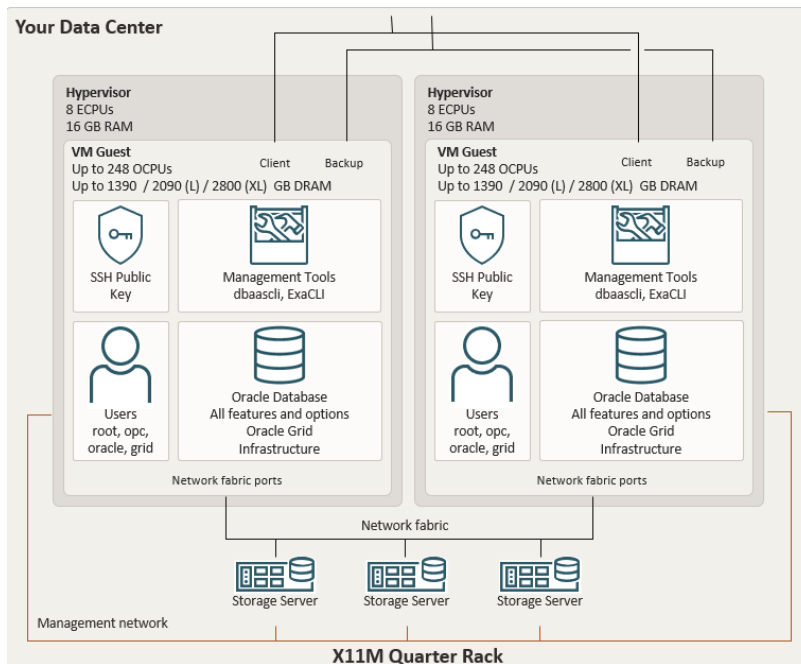
The diagram shows a rack with two database and three storage servers.

On X11M systems, you can host up to 8 VMs per database server.

With Elastic Compute and Storage Expansion, you can dynamically expand the compute capacity with up to 8 database servers and the storage capacity with up to 12 total storage servers.

The client and backup networks provide access to your virtual machine (VM) clusters on the database servers. Oracle manages the infrastructure through the management network, which connects the database and storage server hardware.

3.5.4 Exadata Cloud@Customer VM Clusters and Database Servers



Each Oracle Exadata Cloud@Customer database server contains one or more virtual machine (VM) guests running on a hypervisor and you can designate each VM to host either Exadata Database or Autonomous Database Service instances. This configuration ensures a distinct separation between the Oracle-managed and customer-managed components for the Exadata Cloud@Customer platform.

Oracle manages the hypervisors through the management network. Each hypervisor uses minimal resources: only 2 CPU cores (8 ECPUs) and 16 GB of RAM.

The client and backup networks connect to the VM guests through bonded network interfaces to maximize performance and availability.

If you include Oracle Database software licenses in your Exadata Database Service or Autonomous Database subscription, each VM guest has a complete Oracle Database installation that includes all the features of Oracle Database Enterprise Edition plus all the database enterprise management packs and all the Enterprise Edition options, such as Oracle Database In-Memory and Oracle Real Application Clusters (RAC), as well as Oracle Grid Infrastructure. Alternatively, you can use Oracle Database software licenses that you already own.

Note: Autonomous Database Service prevents customer access to the VM guests because Oracle manages these as part of the service. Administrative actions against the database are limited to protect the service configuration for availability and security. Given that SSH access to VM guests is removed, customer administrative actions in Autonomous Database are performed by an ADMIN user, not by the standard SYSTEM/SYSDBA users, and they're limited to the service-defined APIs, console, and CLI.

Exadata Database Service allows customer access to the VM guests. VM guests require an SSH public/private key pair for operating system security. You register a public key in each guest, and you retain the private key that enables access to the VM operating system. VM guests include standard user accounts, such as oracle, opc, grid, and root.

As a result of this configuration, Exadata Database Service customers manage the VM guests and all the software they contain, including the Oracle-provided management tools, including dbaascli for database lifecycle management and ExaCLI for monitoring and managing your Exadata storage servers.

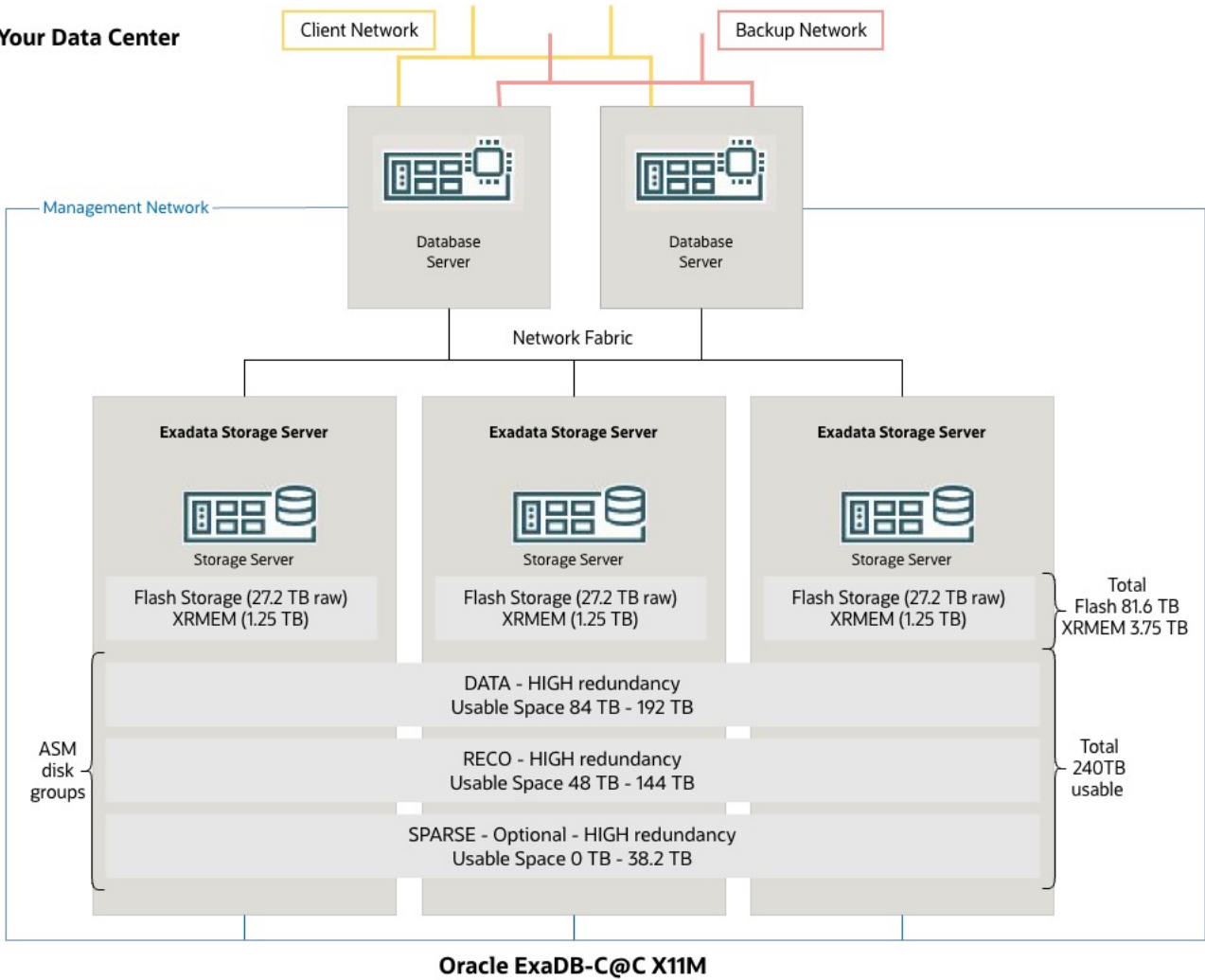
Each X11M database server can have up to 190 cores and 1390 / 2090 (L) / 2800 (XL) GB of DRAM. The diagram shows a quarter rack with two database and three storage servers.

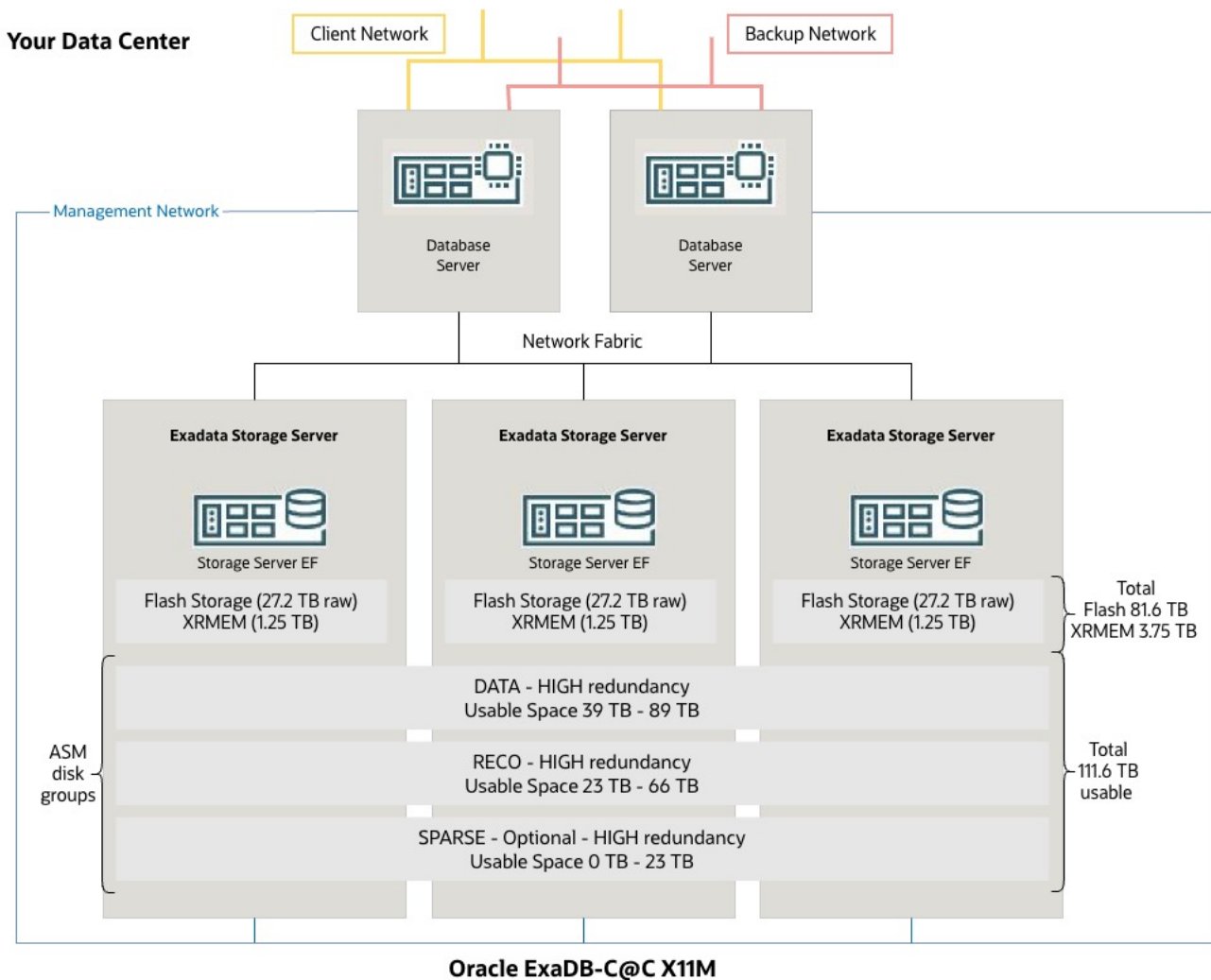
The Exadata Cloud@Customer X11M is also available with a Base configuration, each database server can have up to 30 cores and 660 GB of DRAM.

The database servers connect to the Exadata storage servers through network fabric ports. The Exadata infrastructure resides in your data center.

Oracle manages the infrastructure and, for Autonomous Database, the VM guests and database, through the service-specific management network that connects the database and storage server hardware.

3.5.5 Exadata Cloud@Customer Storage Servers





When you configure Oracle Exadata Cloud@Customer, Oracle Automatic Storage Management (ASM) provisions the storage space inside the Exadata storage servers. By default, ASM creates the following disk groups:

The DATA disk group stores Oracle Database data files.

The RECO disk group stores the Fast Recovery Area (FRA), which is an area of storage where Oracle Database can create and manage various files that are related to backup and recovery.

Only on Exadata Database Service, you can optionally create the SPARSE disk group to support Exadata snapshots. If you create the SPARSE disk group, ASM allocates less space to the DATA and RECO disk groups.

Exadata Cloud@Customer platform customers also need to decide whether to allocate storage to perform local backups to your Exadata storage. If you choose this option, ASM allocates more space to the RECO disk group and less to the DATA disk group.

To learn more about how ASM allocates space to disk groups, see.

In X11 systems, Using the High Capacity (HC) Storage Servers each Exadata storage server has 27.2 TB of raw flash cache and 1.25 TB of XRMEM cache. The usable capacity of a storage servers is 81.6 TB.

Using the Extreme Flash (EF) Storage Servers each Exadata storage server has 27.2 TB of raw flash cache and 1.25 TB of XRMEM cache. The usable capacity of a storage servers is 37.2 TB.

Using the Base Storage Servers each Exadata storage server has 12.8 TB of raw flash cache. The usable capacity of a storage servers is 35.6 TB.

The DATA, RECO and SPARSE ASM disk groups can be created with predefined ratios:

DATA disk group	RECO disk group	SPARSE disk group
80%	20%	0%
40%	60%	0%
60%	20%	20%
35%	50%	15%

The usable storage capacity is the storage available for Oracle Database files after taking into account high-redundancy ASM mirroring (triple mirroring), which provides highly resilient database storage on all Exadata Cloud@Customer configurations.

For Database versions 23ai and 26ai there is an option to use a new storage management architecture called Exascale. Exascale can coexist with ASM on the same storage servers, where the usage of ASM or Exascale can be decided at the creation of the VM clusters.

Exascale can also be used with Database version 19c, 23ai and 26ai to host local filesystems of the VM cluster, which is also an option to select at cluster creation. When chosen the file systems which traditional would be hosted on the local disks in the Database servers will be placed in the Exascale Vaults associated with the VM clusters on the Storage Servers.

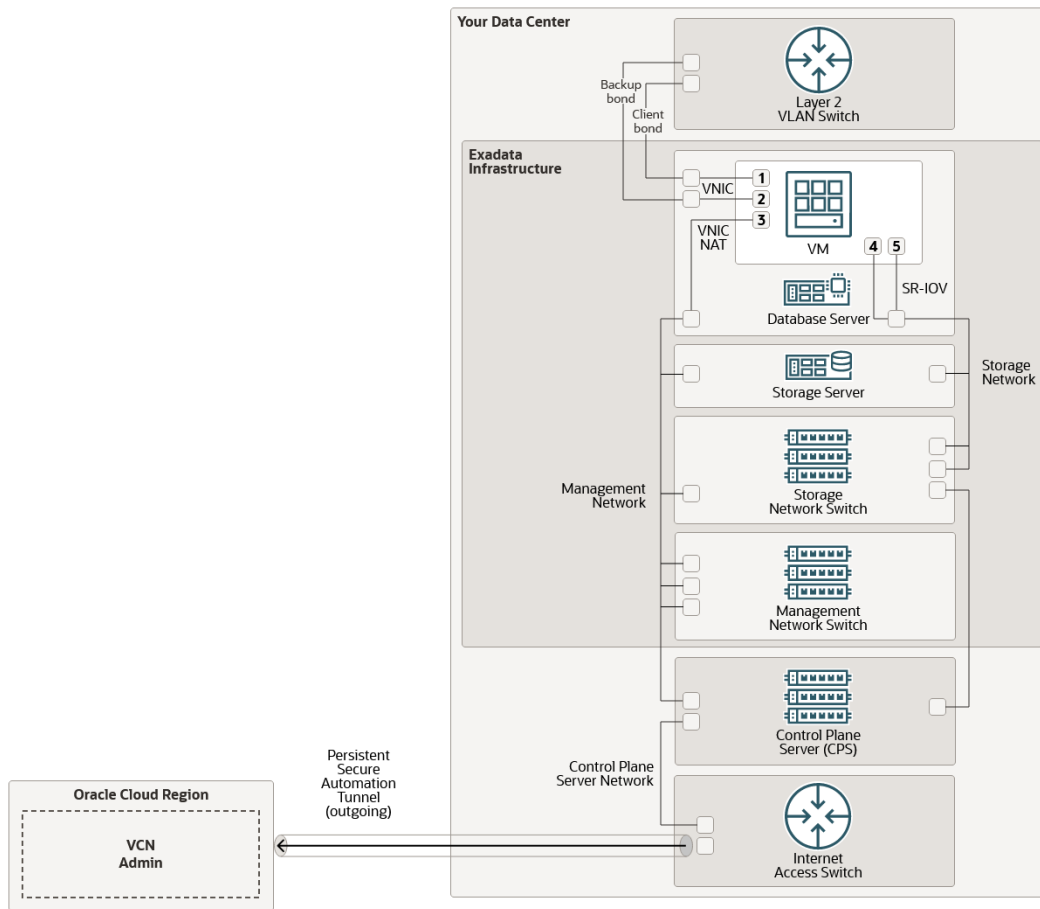
The usable storage capacity does not factor in the use of Exadata compression capabilities, which can increase the effective storage capacity.

The database servers connect to the Exadata storage servers through network fabric ports with active bonding. The Exadata infrastructure resides in your data center. The diagram shows an X11M quarter rack with two database and three storage servers.

Oracle manages the infrastructure through the management network, which connects the database and storage server hardware.

The client and backup networks provide access to your virtual machine (VM) clusters on the database servers. Oracle manages the infrastructure through the management network, which connects the database and storage server hardware.

3.5.6 Exadata Cloud@Customer Network Interfaces



Your databases run on one or more virtual machine (VM) clusters that reside on Exadata infrastructure in your data center. Exadata Cloud@Customer also includes two control plane servers (CPS) that connect to an Oracle Cloud Infrastructure (OCI) region for cloud automation and administration.

Note: For clarity, the diagram displays only one database server, one storage server, and one CPS (control plane server). The client and backup networks provide access to your VM clusters on database servers through a layer 2 virtual local area network (VLAN) switch that you manage and bonded virtual network interface cards (VNICs).

Your VMs access your dedicated Exadata storage servers through a private, nonrouted interconnect network with SR-IOV mapped interfaces. Each physical Exadata database server and storage server has a highly available (HA) (active/standby) connection to a pair of redundant storage networking switches.

The database and storage servers are interconnected through a layer 2 management network and switches. There is no direct access from the management network to your client and backup networks. A subset of Oracle cloud automation functionality accesses your VMs through the management network by using a VNIC interface and a network address translation (NAT) address.

The CPS connects to the storage servers through the storage network switch, and it connects to the management network switch through the control plane network. An outgoing, persistent, secure automation tunnel connects the CPS to the Oracle-managed admin VCN in the OCI region through an internet access switch that you manage.

The following interfaces connect the various networks to the VM:

1. bondeth0 for the client network
2. bondeth1 for the backup network
3. eth0 for the management network
4. re0 for a storage/Real Application Clusters (RAC) interconnect (isolated through VLAN)
5. re1 for a storage/RAC interconnect (isolated through VLAN)

3.5.7 Explanation of Operations Service and RACI model

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

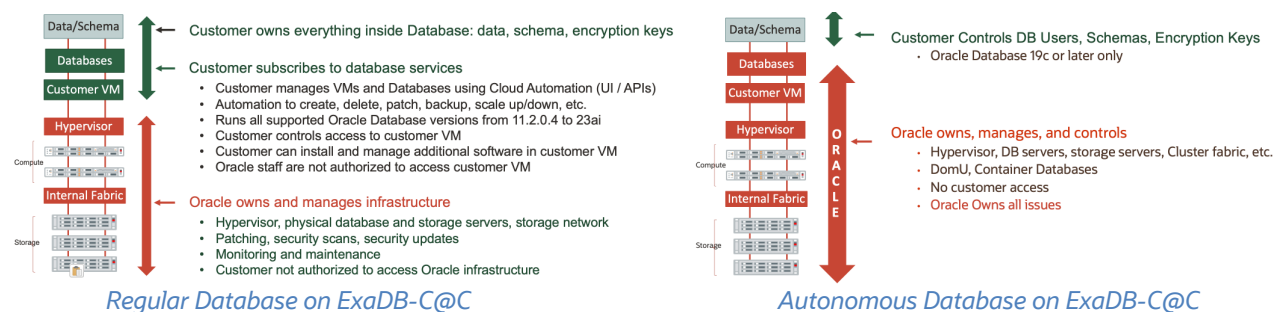
This Explanation of Operations Service describes operational processes for Gen2 ExaDB-C@C including Oracle operations service scope, operations monitoring setup and telemetry data collection, issue identification and remediation, and planned/unplanned maintenance.

Detailed information can be found at following MOS note:

Gen 2 Exadata Database Cloud At Customer - Explanation Of Cloud Operations Service (Doc ID 2707015.1)

The pictures below depict the delineation of the roles and responsibilities between Oracle cloud operations and the end customer using the ExaDB-C@C subscription service. All ExaDB-C@C access is separated and controlled. The customer is solely responsible for maintaining appropriate security, protection, and backup of their content (e.g. applications, databases, data) within their virtual machine (VM). Oracle does not have credentials and does not access the content (applications or data) within the Customer Access Layer. Oracle has complete, exclusive access and administrative responsibilities for the Oracle Infrastructure Layer. The customer is not granted any access or administration privileges within the Oracle Infrastructure Layer.

For more details on security specific roles and responsibilities, as well as access control, please refer to Oracle whitepaper – Oracle Gen 2 Exadata Cloud@Customer Security Controls.



Exadata Cloud at Customer and Exadata Cloud at Customer Autonomous Roles and Responsibilities between Oracle and Customer						
(Disclaimer: Subject to change anytime at Oracle's Discretion)						
Notes:						
(a) (Cloud UI) suggests a functionality is being provided by Cloud UI. In this case, execution of this task is automated and hence will require lot less efforts as compared to manual work.						
		Exadata Cloud at Customer		Cloud at Customer Autonomous		
Component	Tasks	Customer	Oracle	Customer	Oracle	Comments
Database Servers and Software Within Database Servers						
Database Server Hardware	1 Hardware shipment and Installation		X		X	
	2 Upgrades (e.g. Quarter to Half Rack)		X*		X*	*service to upgrade is included in added subscription cost (roadmap item)
	3 Monitor (e.g. Hardware faults)		X		X	
	4 Identification and resolution of issues		X		X	
	5 Incident Management		X		X	
	6 Change Management		X		X	
DomU, GI and Databases	7 Software Installation and Initial Configuration		X		X	
	8 Monitor	X			X	Autonomous - Customer does application performance monitoring, but Oracle monitors for health, uptime, perf regressions
	9 Upgrades (e.g. major Exadata Software Updates. E.g. 12.x to 18.x, 11g to 12c GI and DB updates)	X (Cloud UI)			X	Autonomous - Upgrade automation is a roadmap item, no major upgrades at this time, only latest 19c release available
	10 Tune	X			X	Autonomous - Oracle tunes for the workload, including caching, compression, indexing, and option to use Auto-Indexing and Auto-SPM to tune database - including guarding against regressions. Customer does application performance tuning.
	11 Backup/Restore (DomU OS)	X			X	
	12 Backup/Restore (Databases)	X (Cloud UI)			X	
	12 Identification and resolution of issues	X			X	ExaCC - Customer gathers logs and works with Oracle Support can be engaged via SR. Autonomous - Oracle Ops is first line of defense, does all log capture and analysis and resolves problems without customer if at all possible.
	13 Updates (e.g. minor Exadata Software Updates e.g. 18.1.12 to 18.1.13, quarterly BP for GI and DB)	X (Cloud UI)			X	Autonomous - Customer can optionally set policies to guide autonomous software operations.
	14 Incident Management	X			X	Autonomous - Oracle Ops and Oracle support
	15 Change Management	X			X	Autonomous - customer can set policies for autonomous software operations to follow.
Dom0	16 OS and DB one-off patches	X			X	
	17 Software Installation and Initial Configuration		X		X	
	18 Monitor	X			X	
	19 Upgrades (e.g. major Exadata Software Updates)		X		X	
	20 Tune		X		X	
	21 Backup/Restore (OS)		X		X	
	22 Identification and resolution of issues		X		X	
	23 Updates (e.g. minor Exadata Software Updates)		X		X	
	24 Incident Management		X		X	
	25 Change Management		X		X	
Infrastructure Components Other Than Database Servers	26 OS Security and other Patches		X		X	
	27 Firmware updates		X		X	
	28 Hardware shipment and Installation		X		X	
	29 Upgrades (e.g. Quarter to Half Rack)		X*		X*	service to upgrade is included in added subscription cost (roadmap item)
	30 Monitor (e.g. Hardware faults)		X		X	
	31 Identification and resolution of issues		X		X	
	32 Incident Management		X		X	
	33 Change Management		X		X	
	34 Faulty parts replacements (e.g. disks, flash cards etc)		X		X	
	35 Firmware updates		X		X	
Storage Server Hardware	36 Software Installation and Initial Configuration		X		X	
	37 Monitor		X		X	
	38 Upgrades (e.g. major Exadata Software Updates)		X		X	
	39 Tune		X		X	
	40 Identification and resolution of issues		X		X	
	41 Updates (e.g. minor Exadata Software Updates)		X		X	
	42 Incident Management		X		X	
	43 Change Management		X		X	
Storage Server Software	44 OS Security and other Patches		X		X	
	45 Hardware shipment and Installation		X		X	
	46 Monitor (e.g. Hardware faults)		X		X	
	47 Identification and resolution of issues		X		X	
	48 Incident Management		X		X	
	49 Change Management		X		X	
	50 Firmware updates		X		X	
Networking (1B, Cisco switches) and PDU	51 Hardware shipment and Installation		X		X	
	52 Monitor (e.g. Hardware faults)		X		X	
	53 Identification and resolution of issues		X		X	
	54 Incident Management		X		X	
	55 Change Management		X		X	
	56 Firmware updates		X		X	
	57 Hardware shipment and Installation		X		X	
	58 Monitor (e.g. Hardware faults)		X		X	
Definitions						
Dom0	Initial Domain Started by the Hypervisor on Boot aka administrative domain					
DomU	User Domain					
PDU	Power Distribution Unit					

ExaDB-C@C Roles and Responsibilities

3.5.8 Maintenance and Upgrades

Guide:

This text is generic and should be kept as is, please change the <CUSTOMER NAME>

Example:

Oracle performs updates to all Oracle-managed infrastructure components on Exadata Cloud@Customer. Oracle notifies

specified contacts of infrastructure maintenance by email during infrastructure implementation, These maintenance are defined in a window to determine when quarterly infrastructure maintenance will begin, and view scheduled maintenance runs and maintenance history for your Exadata Cloud@Customer environment in the Oracle Cloud Infrastructure Console. Four maintenance sessions are run each year. The impact during maintenance or possible hardware failures varies depending on how the databases are deployed. The table below provides a summary of unavailability during maintenance.

Component	Owner	Frequency	Comments
Exadata Infrastructure (security update)	Oracle	Monthly (time between patches cannot be more than 6 months)	- Mandatory - Starting date and time at customer's discretion - No database instance restart - Online patching of the DB nodes hypervisor - Rolling patching of the cells
Exadata Infrastructure (full update)	Oracle	Quarterly to bi-annual (time between patches cannot be more than 6 months)	- Mandatory - Starting date and time at customer's discretion - DB nodes are restarted in a rolling fashion
Cluster VM OS	Customer	Quarterly (Recommended)	- The customer decides when to apply the patch - Databases are restarted in a rolling fashion - REST API and dbaascli automate the process
Grid Infrastructure	Customer	Quarterly (Recommended)	
Database	Customer	Quarterly (Recommended)	

- For databases in RAC mode with two active instances, sessions will be redirected to the second active instance.
- For databases in RAC mode with a single active instance/PDB (Singleton Service), sessions will be redirected to the second instance once the service is started on the second instance/CDB

In these first two cases, transactions in progress on the instance that is being interrupted can be replayed if Application Continuity and Transparent Application Fail-Over are active. It should be noted that maintenance starts by stopping application services in drain mode. Without these components, there will be an outage until reconnection on the second instance.

If the database is protected by Data Guard, a switchover can be done in order to perform the upgrade with minimal impact, similar to RAC.

3.6 Sizing and Bill of Materials

Guide:

Estimate and size the physically needed resources of the Workload. The information can be collected and is based upon previously gathered capacities, business user numbers, integration points, or translated existing on-premises resources. The sizing is possibly done with or even without a Physical Architecture. It is okay to make assumptions and to clearly state them! Clarify with sales your assumptions and your sizing. Get your sales to finalize the BoM with discounts or other sales calculations.

3.6.1 Choosing the appropriate sizing method

Guide:

Choose the appropriate sizing approach and method based on your use case

3.6.2 Collected Data for Sizing

Guide:

Available sizing methods:

- *Exadata Cloud@Customer refresh, using the current utilization – section 3.6.3*
- *Oracle Cloud Capacity Analysis (OCCA) – section 3.6.4*
- *Estate Explorer – section 3.6.5*
- *Manual Sizing – section 3.6.6*

3.6.3 Exadata Cloud@Customer refresh data collection

Guide:

In the case of refreshing an existing Exadata Cloud@Customer environment running a sizing exercise is not necessary unless there are new workloads getting introduced into the refreshed infrastructure. In that case another sizing method can be used on its own or in combination with this one. The current resource utilization on the existing Exadata Cloud@Customer has to be captured and mapped in a one-to-one fashion to the new infrastructure. The average OCPU/ECPU consumption can be determined by consulting with the customer, discussing scaling practices and checking the OCI console.

Example:

Current ExaDB-C@C Infrastructure	Number of DB/Storage Servers	Number of cohorts
X9M	2/3	2
X10M-L	4/6	3

Cohort	Allocated Memory	Allocated Storage	Number of DB nodes	Average OCPU consumption for the Cohort
X9M PROD	350 GB	42 TB	2	16
X9M DR	720 GB	90 TB	2	10
X10M-L PROD	720 GB	90 TB	2	40
X10M-L DR	350 GB	42 TB	2	6
X10M-L DEV	480 GB	250 TB	4	48

3.6.4 Oracle Cloud Capacity Analyzis (OCCA) data collection

Guide:

With OCCA Oracle Cloud Capacity Analytics the data collection comes from Oracle Enterprise Manager, AWR (Automatic Workload Repository) and/or is manually added.

Collecting data with Oracle Enterprise Manager is preferable, since the data collection procedure is easier to execute, while the collected data can span up to one year, which helps to identify seasonal trends in the workloads. Revert back to AWR Miner based collection if Enterprise Manager is not used by the customer.

Customer data is gathered with a script provided by Oracle with clear instructions on how to execute. It's customer responsibility to run the extract script and optionally obfuscate the data, and then upload the output to Oracle using secure methods (Oracle provides a Secure Site as part of the ExaDB-C@C sales process).

The output needs a first validation, where any anomalies in the data collection, such as errors, missing data, unrealistic outliers/deviations or unexpected growth can be identified.

This data is then grouped in Cohorts which are in the context of sizing grouping of databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- Network segmentation due to security
- Workload isolation
- Application / User separation
- Required Service levels (for example MAA Gold, Silver, Bronze)
- Size of the workload
- Physical Location

After receiving the data collection files from the customer, follow the OCCA user guide to complete the sizing exercise. Set up the property file with the cohort names for each database based on the discussion with the customer.

Example:

The sizing exercise is based on the following inputs:

Input type	Number of Database	Time span covered
AWR Miner	8	06/01/2025 - 06/30/2025
EMCC Extract	34	06/30/2024 - 06/30/2025

3.6.5 Sizing with Estate Explorer

Guide:

Oracle Estate Explorer is a free analysis tool available to customers, that reveals operational and ownership insights, TCO alternatives, and migration readiness across their Oracle Database estate. Oracle Estate Explorer runs a series of metadata queries against the data dictionary - all from within a single anonymous pl/sql block. The queries are available for the customer to check, and the output is a CSV file with embedded json - again which the customer can check before sharing with Oracle.

Oracle Estate Explorer provides enough metrics to facilitate a high level sizing or 'not to exceed' sizing, which depending on the customer scenario, in some cases is good enough and is much quicker and easier to conduct than a detailed sizing exercise with tools such as OCCA.

Oracle Estate Explore is ideal when assessing 10 Databases or more and the business is open to using ADB. The databases under assessment can be anywhere - on prem, OCI or a competitor cloud. Ideally, the customer is using Enterprise Manager to manage the estate or a list of databases with connection details is available.

The Oracle Estate Explorer is an APEX app, so it can be installed anywhere you can run APEX.

Customers who are sensitive to sharing their information with Oracle often install OEE on-premises, and that can be in any Oracle 19c database with the latest version of APEX - this can be on-premises, in a VM, in OCI, in a competitor cloud and of course in ADB in a customer tenancy (including free tier).

For those customers who are willing to share their information with Oracle, then pre-sales can host an instance internally and analyze the data on customer's behalf.

Example:

Datacenter	Database Estate	Number of Databases
Primary	Production OLTP	27
Primary	DWH	3
Secondary	Production OLTP DR	27
Secondary	DWH DR	3
Secondary	Test/Dev	20

3.6.6 Manual sizing

Guide:

Wherever possible we try to avoid manual sizing efforts. Since the data input is limited, that can affect the correctness of the output and. This method can only be used as a not-to-exceed type sizing, with no workload seasonality, which usually means that from all the sizing methods manual sizing can produce the least accurate results. Whenever it is possible please use a different sizing method.

Manual data collection can mean several things, for example:

- Finding the environments in scope, for example PROD, TEST, DEV, QA, etc*
- Understanding the number of databases*
- Versions used*
- RAC or Single Instance*
- Dataguard or not*
- Looking at number of licenses in use by the customer*
- Collecting data on existing hardware platform by interviewing customer*
- Asking for customer estimates and/or metrics on CPU, memory and storage utilization in current environment*

- *Using backup data to estimate size of databases (querying Full Backup size from RMAN catalog)*

Statspack collection based sizing should be only used in an effort to determine the workload of Standard Edition Database environments, which we want to move to an Exadata Cloud@Customer system.

Example:

The sizing exercise is based on the following inputs:

Infrastructure	Number of Databases	CPU type	Allocated number of cores	Allocated Memory	Allocated Storage
Oracle Exadata X6-2	8	Intel Xeon E5-2699-v4	16	650 GB	24 TB
Generic x86	28	Inter Xeon Gold 6328HL	42	1390 GB	98 TB
IBM Power	10	POWER9 SMT8	20	380 GB	40 TB

!

3.6.7 Defined VM Clusters for Sizing

Guide:

Cohorts in the context of sizing are grouping databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- *Network segmentation due to security*
- *Workload isolation*
- *Application / User separation*
- *Required Service levels (for example MAA Gold, Silver, Bronze)*
- *Size of the workload*
- *Physical Location*

Document the VM Clusters that has been agreed with the customer and any assumptions made. If you have a more complex VM Cluster layout with VM Cluster Node Subsetting ([Overview of VM Cluster Node Subsetting](#)) then consider visualizing it. There is an example below.

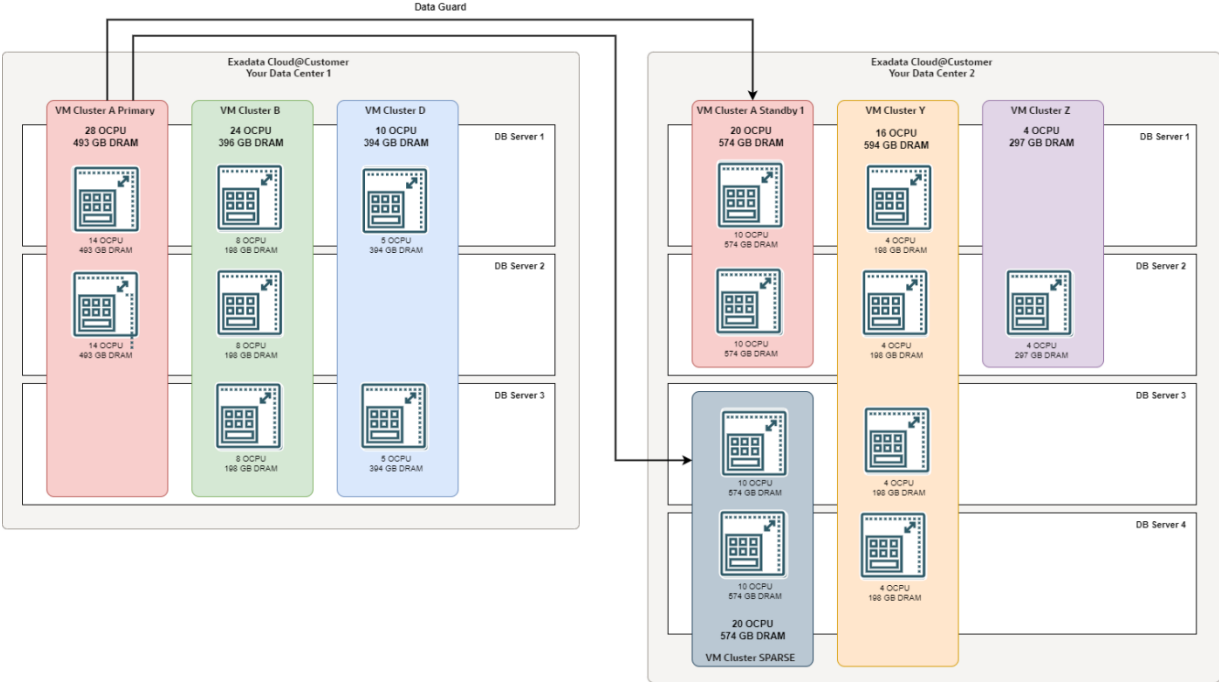


Figure 1 - Complex VM Cluster layout visualization

Example:
Based on the customers' two different data centers and requirements for Production and non-Production environments which include different service levels as well as separate Client network segments for the environments, the following Cohorts (VM Clusters) have been defined.
Each VM cluster runs a single version of the Oracle Grid infrastructure (GI) to be chosen by the customer among the most recent. Each cluster can run different versions of Oracle Homes. Each independent VM clusters has ECPU assigned during the deploy phase of the VM cluster, so, we can assign different number of ECPU to different VM clusters and environments. VM Clusters to be deployed are listed in the following tables together with the criteria that define why separate VM Clusters are deployed.

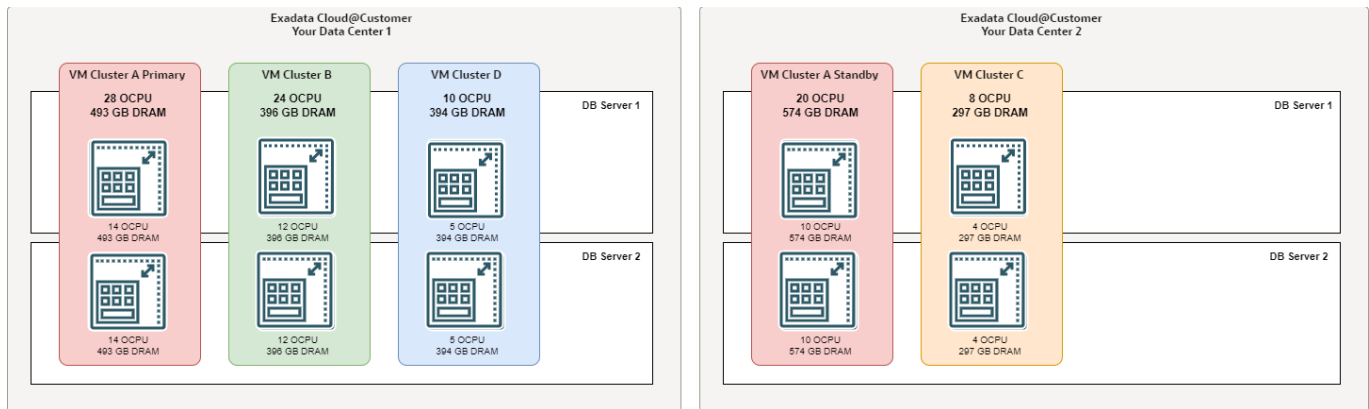
Your Data Center 1

Data Center 1 <NICKNAME > VM Clusters	Environment	Client Network Segment	Backup Network Segment	MAA Service Level
Production A Primary	Production	Client Network A	Backup Network	Gold
Non-Production B	Test	Client Network B	Backup Network	Silver
Miscellaneous D	Test	Client Network D	Backup Network	Silver

Your Data Center 2

Data Center 2 <NICKNAME > VM Clusters	Environment	Client Network Segment	Backup Network Segment	MAA Service Level
Production A Standby	Production	Client Network A	Backup Network	Gold
Non-Production C	Test	Client Network C	Backup Network	Silver

The following illustrates the placement of the five VM Clusters on the two Exadata Cloud@Customer systems and the VM's created on each DB Server as part of the VM Clusters.



An ExaDB-C@C X11M Quarter Rack (with two DB Servers) can host up to 12 independent VM Clusters, the maximum number of VM Clusters is 24 for X11M systems with more than 2 database servers. The maximum number of VMs per database server is 8 regardless of the number of database servers in the system.

Please see the ExaDB-C@C X11M Data Sheet for the latest information: [Exadata Cloud@Customer X11M Data Sheet](#)
 Documentation on Managing VM Clusters in Exadata Cloud@Customer: [Overview of VM Cluster Node Subsetting](#)

3.6.8 Sizing Assumptions

Guide:

Document the customer requirements that is included in the sizing as well as assumptions made for the sizing. This information is provided as part of the sizing exercise.

Scaling

If the data collection captured performance requirements over time (OCCA-based sizing), the impact of scaling can be calculated in the results. In that case, indicate the time utilization percentages used (e.g., 90%, 85%, 80%) in the results.

Growth

The predicted growth of the targeted workloads needs to be taken into account to ensure that the customer has sufficient infrastructure available, and is aware of the increase of ECPU consumption over time.

This growth will be composed of an assumption of the expected organic growth of the existing target workloads, but additional new or not currently in scope workloads should also be considered.

The ramp up plan of ECPU consumption should take into account the on boarding plan of the workloads for the customer. e.g. They may start with a smaller test environment prior to starting migrations and moving into production.

DR environment sizing

Under normal operating conditions, pure DR and readable Standby databases will have lower processing demands than during a disaster. During normal operations, required vCPU and I/O consumption will typically be lower.

Single Instance to RAC conversions

When moving from Single Instance DB to a 2-node RAC DB, it's advised to allocate the same amount of memory that would be allocated to the single VM to both nodes in the VM-Cluster. Hence doubling the amount of memory. When in this case a node would go down, the remaining node can run the DB activities with no performance impact.

SGA sizing – Large Pages

SGAs are often over sized. The AWR report for specific databases i.e. the SGA and PGA advisor can determine whether it is safe to reduce. For dedicated platform cloud deployments (ExaDB-C@C, ExaDB-D, ExaDB-D@Hyperscaler) it is also important to consider settings for Large Pages.

When a VM is initially created, each VM's operating system is configured with 50% of the memory allocated to the VM for large pages, and databases are configured to use that memory for their SGA. Oracle recommends that you not modify the large pages configuration unless you understand the implication of any changes you make. If you configure large pages to be greater than 60% of total memory, then the cloud automation will resize it to 60% of total memory. This automatic resize is to ensure sufficient conventional memory for the virtual machine to start.

Storage sizing

Applying or introducing compression can have a significant impact on the required storage. The compression ratio depends highly on the data-set and the selected compression algorithm. Applying compression often needs some thought work from

the customer and sometimes will only be applied after DB migration. Make sure there is sufficient buffer capacity in the target platform.

It's advised to use Compression Advisor to estimate any compression ratio's applied during a sizing exercise. Document and share any assumption taken.

With the introduction of Exascale technology storage management for Exadata deployment options has changed. The need to have ASM diskgroups with allocated storage capacity is no longer there. For Exa dedicated Cloud deployments models (ExaDB-C@C, ExaDB-D in OCI and ExaDB-D@Hyperscaler) the fixed ratio's for DATA/RECO or DATA/RECO/SPARSE is removed with Exascale. On the other hand, if customer desires to implement cloning with Exascale there is storage capacity to reserve for this purpose.

The following example has some useful assumptions that should be considered.

Example:

The following customer requirements and general assumptions have been used in the sizing.

- All analyses are based on previous resource consumption and does not necessarily predict the future workload behavior.
- The defined cohorts (VM Clusters) and databases can differ from the current deployment.
- The sizing computations are based on the hourly average resource consumption of databases in each cohort.
- Cohort metrics are aligned by time and then the sum of peaks is computed for each hour.
- With OCCA based sizing the utilization percentage is indicated (90/85/80%)
- Hourly data is adjusted based on peaks found in the daily data over the past year (valid for EM data case).
- Using in-memory or other database options may reduce/increase the required processing power.
- Using different execution plans will influence the required processing power.
- For each VM Cluster, an additional 16 ECPU's were added for OS+GI. Metrics for DR DBs were considered as 20% of the corresponding Primary DBs

Infrastructure growth

The solution needs to be sized to accommodate 4 or 5 years of growth. For other Oracle Database Services, the chosen platform needs to be capable of expanding to accommodate future growth for a reasonable amount of time before re-platforming becomes necessary.

Sizing Attribute	Overall Growth Percentage
CPU Growth %	xx %
Memory Growth %	xx %
Storage Growth %	xx %
Storage IOPS Growth %	xx %
Sizing Outlook Years	4/5

Next to the infrastructure growth percentages it is also wise to make sure there is a safety margin and not fill-up infrastructure for the full 100% of what is physically available/installed.

Sizing Attribute	Margin Percentage
CPU margin %	xx %
Memory margin %	xx %
Storage margin %	xx %

DR environment sizing

Standby Sizing Attribute	Normal Operation (Standby Mode)	Activated During Disaster
Database Storage	100 % of Primary	100 % of Primary

Standby Sizing Attribute	Normal Operation (Standby Mode)	Activated During Disaster
Instance Memory	100 % of Primary	100 % of Primary
Sessions	100 % of Primary	100 % of Primary
ECPU consumption	N % of Primary (default 20 %)	100 % of Primary
I/O consumption	N % of Primary (default 20 %)	100 % of Primary

Single Instance to RAC conversion

Sizing Attribute	Margin Percentage
CPU Overhead % (Overhead due to RAC cache fusion)	xx %
RAC Memory Overhead % (Overhead to handle node failures)	xx %

SGA sizing – Large Pages

Sizing Attribute	Margin Percentage
Large Pages Memory Alloc.(50% default, 60% max)	xx %

Storage sizing

Sizing Attribute	Margin Percentage
Percentage of overall storage capacity or data files (DATA)	xx %
Percentage of overall storage capacity for redo, fra, ... (RECO)	xx %
Percentage of overall storage capacity for cloning (space to keep older data blocks)	xx %

3.6.9 Proposed Hardware Shapes*Guide:*

Document the proposed hardware shape and any additions, corrections or customer requested changes made.

Database or Storage Server expansions: list the details of the new hardware needed by the customer. Since it is important to get an overview of the ExaDB-C@C configuration after including the new HWs, include also a description of the overall configuration.

If the expansion involve Database Servers, update the Consumption Plan.

Example:

With the summarized sizing information applied to High Availability and Performance Best Practices both in general and specific to the Exadata Cloud@Customer, the following hardware shapes are recommended.

Datacenter	Infrastructure	Number of DB Servers	Number of Storage Servers	Number of Expansion Racks
Primary	ExaDB-C@C X11M-L	4	16	1

Datacenter	Infrastructure	Number of DB Servers	Number of Storage Servers	Number of Expansion Racks
Secondary	ExaDB-C@C X11M-L	4	6	0

For more information and specifications on the Exadata Cloud@Customer X11M system, please see the data sheet at this link: [Exadata Cloud@Customer X11M Data Sheet](#)

3.6.10 Estimated Consumption Plan

Guide:

Exadata Cloud@Customer infrastructure runs over a 48-month commitment with the possibility for the customer to extend an additional 12 months for a total of 60 months or 5 years.

The consumption can either be covered by BYOL, Bring Your Own Licenses, or by Licenses Included. With BYOL customer will consume BYOL ECPU and will also be required to have sufficient licenses for DB and Options (except TDE which is included). Choice of BYOL or Licenses included are per VM Cluster.

Using customer requirements and assumptions made, document the estimated ECPU consumption over time.

Example:

This is the proposed Consumption Plan over five years based on the sizing, growth and other assumptions including OS/GI load.

Datacenter	Cohort	Year 1 ECPU per hour	Year 2 ECPU per hour	Year 3 ECPU per hour	Year 4 ECPU per hour	Year 5 ECPU per hour
Primary	PROD	20	22	26	32	38
Secondary	DR	4	6	6	6	8
Secondary	TEST/DEV	12	14	16	18	20
Total:		36	42	48	56	66

3.6.11 Commercial & Technical Exceptions

Guide:

This is internal section for reference. Delete before creating customer version.

Data Centre	Exception Details
Data Center 1 <NICKNAME>	
Data Center 2 <NICKNAME>	

3.6.12 Bill of Materials

Guide:

You can add the Bill of Materials in any shape or format. Create a table, bullet list or a picture from the CPQ quote.

Exadata Cloud@Customer infrastructure needs to have a CPQ quote for each location it is going to be shipped to (due to Ship-to addresses and Site Details in the SAR Service Activation Request potentially being different). So, for a solution with two data centers there will be a minimum of two CPQ quotes, hence you should list that as two independent BOM.

With the Exadata Cloud@Customer X11M generation the Rack infrastructure has two Database Servers included and NO Storage servers included. A minimum of three Storage Servers has to be included with each configuration. The Exadata Cloud@Customer consumption can be either based on BYOL or License Included, or both (VM Cluster level). For example, VM Cluster 1 can be License Included while VM Cluster 2 can be BYOL. Correct SKU's need to be used and can be added to one of the BOM if there are several.

Example:

This is the Bill of Material (BOM) for the two data centers. The ECPU consumption for License Included is added on one of the BOM's.

Your Data Center 1 - Bill of Material line items	Quantity	Term
Exadata Cloud@Customer - Rack - X11M - Hosted Environment Per Month	1	48 months
Exadata Cloud@Customer - Storage Server - X11M - Hosted Environment Per Month	3	48 months
Gen 2 Exadata Cloud at Customer Installation and Activation Service	1	n/a
Oracle Exadata Cloud@Customer Database ECPU - ECPU Per Hour	328	12 months

Your Data Center 2 - Bill of Material line items	Quantity	Term
Exadata Cloud@Customer - Rack - X11M - Hosted Environment Per Month	1	48 months
Exadata Cloud@Customer - Storage Server - X11M - Hosted Environment Per Month	3	48 months
Gen 2 Exadata Cloud at Customer Installation and Activation Service	1	n/a

Annex

4.1 Security Guidelines

4.1.1 Oracle Security, Identity, and Compliance

Oracle Cloud Infrastructure (OCI) is designed to protect customer workloads with a security-first approach across compute, network, and storage – down to the hardware. It's complemented by essential security services to provide the required levels of security for your most business-critical workloads.

- [Security Strategy](#) – To create a successful security strategy and architecture for your deployments on OCI, it's helpful to understand Oracle's security principles and the OCI security services landscape.
- The [security pillar capabilities](#) reflect fundamental security principles for architecture, deployment, and maintenance. The best practices in the security pillar, help your organization to define a secure cloud architecture, identify and implement the right security controls, and monitor and prevent issues such as configuration drift.

4.1.2 References

- The Best Practices Framework for OCI provides architectural guidance about how to build OCI services in a secure fashion, based on recommendations in the [Best practices framework for Oracle Cloud Infrastructure](#).
- Learn more about [Oracle Cloud Security Practices](#).
- For detailed information about security responsibilities in Oracle Cloud Infrastructure, see the [Oracle Cloud Infrastructure Security Guide](#).

4.1.3 Compliance and Regulations

Cloud computing is fundamentally different from traditional on-premises computing. In the traditional model, organizations are typically in full control of their technology infrastructure located on-premises (e.g., physical control of the hardware, and full control over the technology stack in production). In the cloud, organizations leverage resources and practices that are

under the control of the cloud service provider, while still retaining some control and responsibility over other components of their IT solution. As a result, managing security and privacy in the cloud is often a shared responsibility between the cloud customer and the cloud service provider. The distribution of responsibilities between the cloud service provider and the customer also varies based on the nature of the cloud service (IaaS, PaaS, SaaS).

4.1.4 Additional Resources

- [Oracle Cloud Compliance](#) – Oracle is committed to helping customers operate globally in a fast-changing business environment and address the challenges of an ever more complex regulatory environment. This site is a primary reference for customers on the Shared Management Model with Attestations and Advisories.
- [Oracle Security Practices](#) – Oracle's security practices are multidimensional, encompassing how the company develops and manages enterprise systems, and cloud and on-premises products and services.
- [Oracle Cloud Security Practices](#) documents.
- [Contract Documents](#) for Oracle Cloud Services.
- [OCI Shared Security Model](#)
- [OCI Cloud Adoption Framework Security Strategy](#)
- [OCI Security Guide](#)
- [OCI Cloud Adoption Framework Security chapter](#)

4.1.5 Administration Annex

Service/Product Name	Description	Collateral
OCI Console	The OCI Console is embedded in all cloud services. It allows basic tasks such as listing, starting, stopping, or termination of resources.	OCI Console
OCI Database Management	This OCI-managed service allows you to manage your databases. It provides a subset of functionalities offered by the OEM.	Database Management
OCI Organization Management	The OCI Console has several tenancy management features. You can use Organization Management to centrally manage your multi-tenancy environment.	Organization Management

4.1.6 Troubleshooting Annex

Service/Product Name	Description	Collateral
Logging Analytics	OCI Logging Analytics can handle log events generated by all software applications and infrastructure on the cloud or on-premises. For Oracle software logs, a predefined severity pre-classification exists based on Oracle experience.	OCI Logging Analytics OCI Logging Analytics for Exa
OCI Application Performance Monitor	APM allows to drill down from user sessions till the application logs to find the root cause.	OCI Application Performance Monitoring
OCI Database Management	OCI-managed service that allows you to drill down and correlate metrics and data from different layers. It provides built-in links that allow you to connect to other O&M services (ex. Ops Insights).	Database Management
Ops Insights	OCI Ops Insights allows tracking of metrics charts and data collection. It allows for the correlation of resources from different infrastructure layers.	OCI Operations Insight OCI ExaInsight

4.1.7 Cost Control and Chargeback Annex

Service/Product Name	Description	Collateral
Ops Insights Capacity Planning	This OCI-managed service allows one to predict the resource consumption for a year. With tags, you can associate the forecast and the consumption to a specific department.	Operations Insight Capacity Planning
Cost Analysis	Cost Analysis is an easy-to-use visualization tool to help you track and optimize your Oracle Cloud Infrastructure spending. It allows for the generation of charts and the download of accurate and reliable tabular reports of aggregated cost data. With tags, you can associate the forecast and the consumption to a specific department.	OCI Cost Analysis
Usage RestAPI	OCI offers various RestAPI's to manage services, including the one for cost management.	OCI Usage RestAPI

4.2 Maintenance and Patching

The following guidelines are recommended to reduce the operational risk and business impact during customer maintenance activities:

- Schedule updates during low-usage periods and outside business hours to minimize disruption. Ensure that patching activities across different components are coordinated to avoid scheduling conflicts.
- Whenever it is possible, execute the patching activities in this order: Exadata Infra → VM Cluster → Oracle Grid Infrastructure → Oracle Database Software → Container Database and add buffers between maintenances in case of unexpected issues.
- Always schedule non-production before production environments and follow a layered approach per environment. This enables your application teams to run functional and performance tests to identify any potential regressions before they may impact the business and to correlate any possible issue with the corresponding patching event.
- Ensure that all components scheduled for patching are operational and available before starting the update.
- Before you apply any patch, run the precheck operation to ensure your VM cluster, Grid Infrastructure, or Database Home meets the requirements for that patch.
- Maintaining system security and performance requires **regular patching**, including monthly OS updates and quarterly database and Grid Infrastructure patches.
- Oracle recommends your Oracle Grid Infrastructure RU version be no more than 6 months behind your latest database RU version. When updating the database version, you should also update the GI version if possible.
- Use OCI-supported **Software Images**, including custom database and Grid Infrastructure images with RU and interim patches, for provisioning and patching.
- Installing OS packages outside of Oracle's official Exadata updates can create dependency issues that disrupt future updates, so Oracle advises minimizing such customizations. If you choose to install extra packages, it's best to use scripts that automate their removal and reinstallation to avoid complications during updates.
- You must use **cloud-native interfaces** like OCI Console, API, CLI, **dbaascli**, or fleet update services, instead of manual tools like opatch, unless explicitly indicated.
- Ensure you have a valid backup of your databases before you apply any patches.
- Oracle recommends out-of-place patching for databases and Grid Infrastructure, that implies moving them to a Home that uses the target patching level when possible. This technique requires less downtime and allows you to easily roll back to the previous version by moving it back to the old Home.
- Additional Oracle tools, such as Oracle Real Application Testing (RAT), can be utilized to support performance tests and replicate production database load in production-like environments.
- In mission-critical databases protected by Oracle Data Guard, standby databases are always patched first. You could convert the already patched physical standby database into a snapshot standby database and test the patch before the primary is applied. This step would reduce the database protection against disaster temporarily while the tests are run. The risk can be mitigated with the use of a second standby database. The recommendation is to convert one standby to a snapshot standby and leave the other standby for protection purposes only.

- When managing updates across a large number of guest VMs, Grid Infrastructure or database deployments, it is recommended to leverage fleet maintenance capabilities, such as Exadata Fleet Update or Fleet Application Management, to streamline patching, ensure consistency, and reduce operational overhead.
- Patch activities can be monitored through the OCI Console by reviewing Work Request logs or the logs of the fleet services used for the updates. For guest VMs, Grid Infrastructure, and databases, the default component-specific logs are also available to track progress and identify issues.
- Subscribe to maintenance events to get reminders and status on maintenance activities, for more details refer to the information provided in [Oracle Exadata Database Service on Cloud@Customer Events](#).
- For additional reference you can check the public, latest documentation on [Configure Oracle-Managed Infrastructure Maintenance & Patch and Update an Oracle Exadata Database Service on Cloud@Customer System](#).